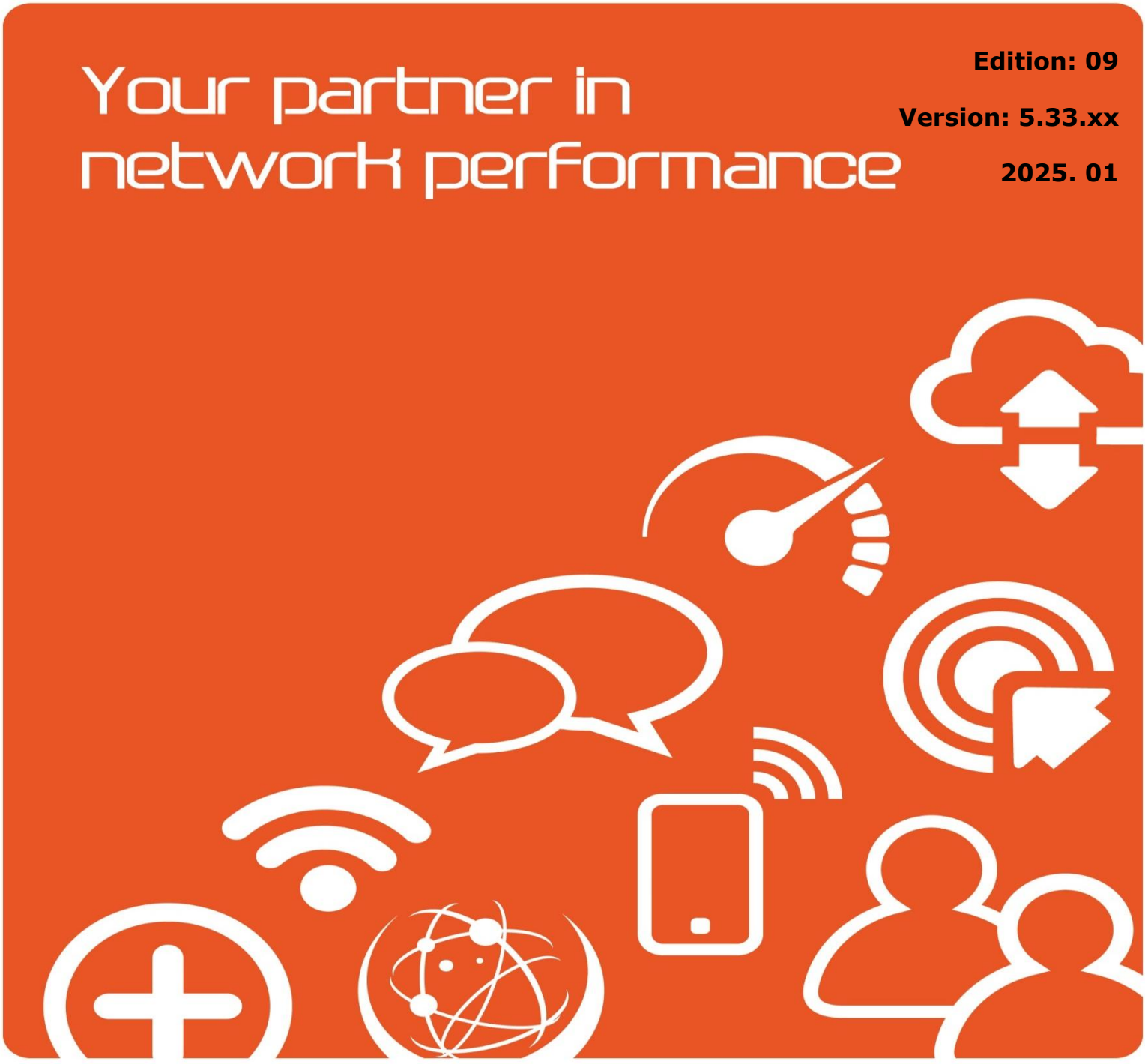
XCAP
User Guide

Your partner in network performance

Edition: 09

Version: 5.33.xx

2025. 01



© 2025 Accuver Co., Ltd., All rights reserved.

This documentation shall not be changed, distributed to the public, and opened to the third person without written permission. Accuver is not responsible for any direct or indirect damages arisen or related to use of this manual. All information included herein may be changed without prior notice.

Accuver Co., Ltd.,

Address: Innowireless B/D 190, Seohyeon-ro, Bundang-gu, Seongnam-si, Gyeonggi-do, South Korea

Web site: www.accuver.com www.innowireless.com

Phone #: +82-31-788-1700

Fax #: +82-31-705-1246

Revision History

Revision	Product Version	Date	Description
01	5.22.xx	Mar, 2019	Initial Version
02	5.24.xx	Jan, 2020	
03	5.26.xx	Mar, 2021	
04	5.28.xx	Jan, 2022	
05	5.30.xx	Jan, 2023	
06	5.31.xx	Sep, 2023	
07	5.31.xx	Nov, 2023	
08	5.32.xx	Apr, 2024	
09	5.33.xx	Jan, 2025	

Contents

Revision History **3**
Contents 4
Scope **6**
Overview 6
Key Functions and Features 7
Functions 7
Features 8
License Policy 9
System Requirements 10
Installation **11**
Installing Software (with Setup Package) 11
Getting Started **12**
Executing XCAP Program 12
General Analysis Procedure 12
Creating New Model (Parsing Log Data) 14
Opening Model 18
Parameter Window Open 20
Opening Graph Window 21
Opening Map Window 23
Opening Table Window 27
Opening CDF/PDF Graph Window 30
Opening Event Window 31
Opening Statistics Window 32
Opening Message Window 33
Analysis Method **36**
Basic Menu **38**
Main Window 38
Main Menu 38

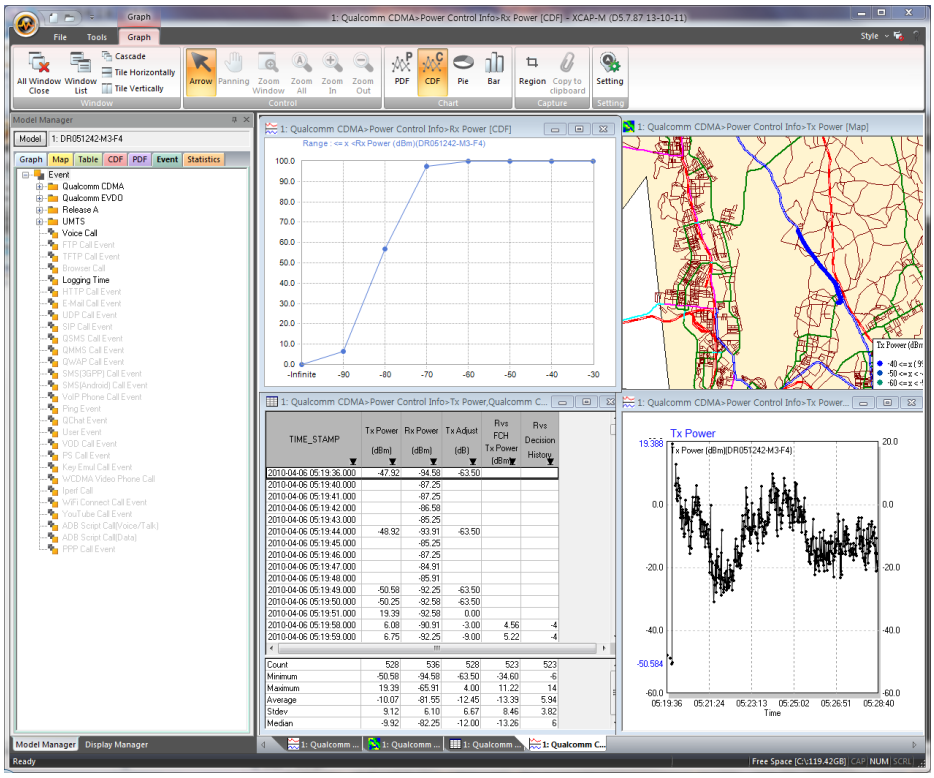
File 39
Tools 41
Window Bar 43
Work Bar 48
Window Mouse Menu 52
Tree Mouse Menu 53
Contact Us 54

Scope

Overview

XCAP is post-processing software that analyzes logging data for IS-95A/B, CDMA2000, EV-DO, WCDMA, GSM, LTE, TD-SCDMA, NB-IoT and 5G NR collected from diagnostics monitoring devices. It offers full featured tools to optimize network maintained and troubleshot.

XCAP presents the data in various GUI types; Graph, Map, Table, Message, CDF, and PDF. By simply dragging and dropping parameters, one or multiple parameters are shown in one parameter window. It displays data using customized filtering tools as well as generates various reports automatically.



Key Functions and Features

Functions

XCAP is provided with the following powerful functionalities.

Graph/Map analysis

- Zoom in/out, range zoom
- Multi-layer overlap
- Correlation graph
- CDF/PDF graph
- Sync, panning, show average, export, etc.

Table analysis

- Show data in time order.
- Import/export data.
- Organize user-selected tables.
- Display all data in one table analysis window.

Import

- Import and analyze CSV, txt, Tektronix, Ethereal, TEMS.
- Analyze imported files using Graph, Map, Table, CDF, PDF.

Filtering

- Display specific data with filter.
- Apply filtered data to all parameter windows.

Replay

- Replay data in time order.

Auto reporting

- Export various reports automatically.
- Export to Excel and PowerPoint.

© Accuver Co., Ltd.

Features

XCAP includes special features.

File merge

You may analyze several logging files as one file by using file merge function.

Various analysis forms

XCAP offers various analysis types such as Graph, Map, Message, Table, CDF, and PDF.

Powerful mouse menu

XCAP utilizes a powerful pop-up menu for effective analysis on every parameter window. Simply right-click on every parameter window.

Data export

Analyzed data in XCAP can be exported to Excel, PowerPoint, CSV, Txt, and etc. Saving formats may vary depending on analysis types.

Favorite organization

For Create Template function: You may organize parameter windows in different analysis types, and reload from Message tab.

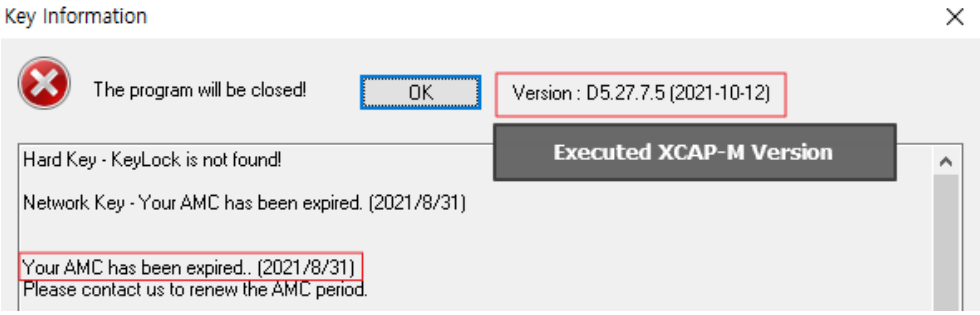
For Create Favorite function: You may organize parameter windows in a same analysis type, and reload from the selected analysis type tab.

Multi window synchronization

XCAP allows you to analyze multiple parameter windows simultaneously by using Range to Sync function.

License Policy

We have an AMC Annual Maintenance Contract, and you can receive the latest patch for free within one year from the date of signing of the AMC. Programs released before the AMC expiration date may be used regardless of the AMC expiration date.



Notification window when AMC expires

System Requirements

Before you begin, ensure that your system meets the following requirements.

Item	Minimum Requirements	Recommended Requirements
CPU	Core2 Duo Processor, 2GHz	Quad Core (Thread Count: 8) processor, 2.4 GHz or above
Monitor Resolution	1680 * 1050(WSXGA+) or above	1920 * 1080(Full HD)
RAM	8GB or above	16GB
Hard Drive	1TB or larger hard disk for collecting data	10TB (7200 RPM) or larger hard disk for collecting data Recommendation: SSD
Operation System	Windows 10 or higher OS	Windows 11

Installation

Installing Software (with Setup Package)

Following procedure enables you to install XCAP on the computer appropriately. All applications should be closed on your PC before beginning installation.

1. Insert the supplied CD-ROM into your CD-ROM drive.

   The CD-ROM shall automatically run, and an installation window appears.

   If the installation window does not appear, find the **setup.exe** from the supplied CD-ROM/XCAP Setup.

2. Installation wizard is started, read carefully and follow installation instructions properly.
3. Press **Finish**.

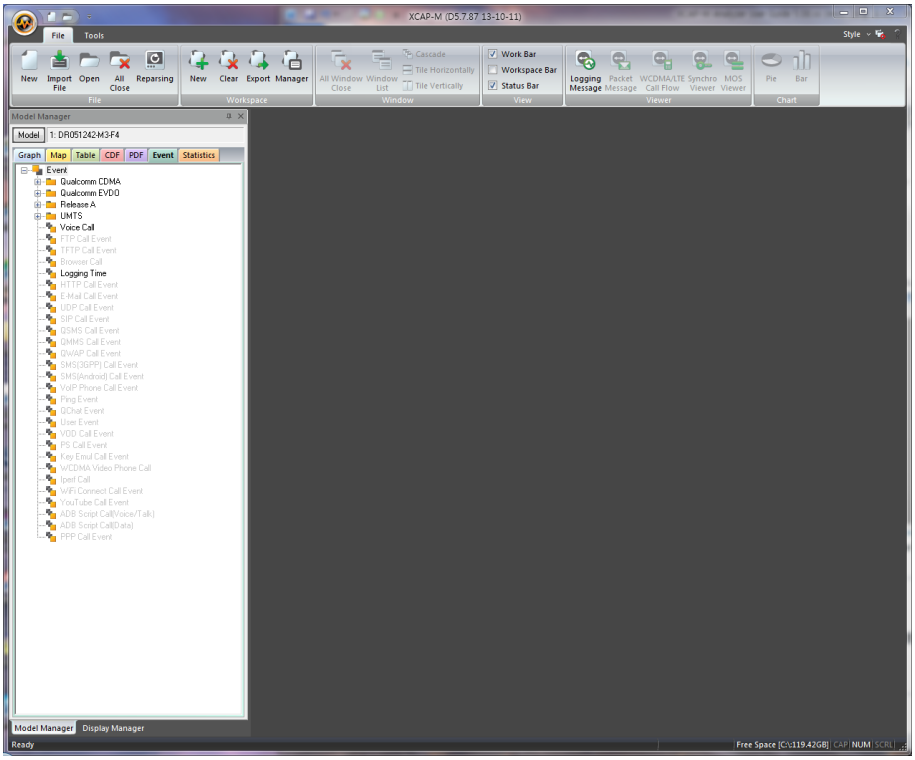
Getting Started

Now, all necessary installation steps before starting is completed and ready to use XCAP program. This chapter explains how to execute XCAP program in PC, and use basic options and features to analyze voice/data call, network quality data and etc. in various formats.

Executing XCAP Program

1. Click XCAP icon 

in Desktop to execute XCAP program.
2. XCAP main window appears.



XCAP Main Window

General Analysis Procedure

	1. Create Model by parsing log data. ✓ For details, see **Creating New Model (Parsing Log Data)**. **2. Filter Call Events to analyze in Call Statistics window.**

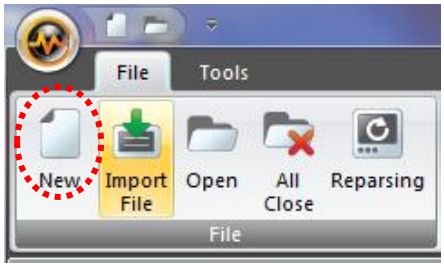
	✓ For details, see **Opening Statistics window**. ✓ For more information, see the Filtering section of the accompanying documentation. **3. Synchronize Call Events in various analysis windows.** ✓ For more information, see the Range to Sync section of the accompanying documentation. ✓ For more information, see the Analysis Method section of the accompanying documentation. **4. Setting detailed analysis options.** ✓ For more information, see the Tools section of the accompanying documentation. **5. Publish report in various formats.** ✓ For details, see Menu Bar – **Report Menu**.

Creating New Model (Parsing Log Data)

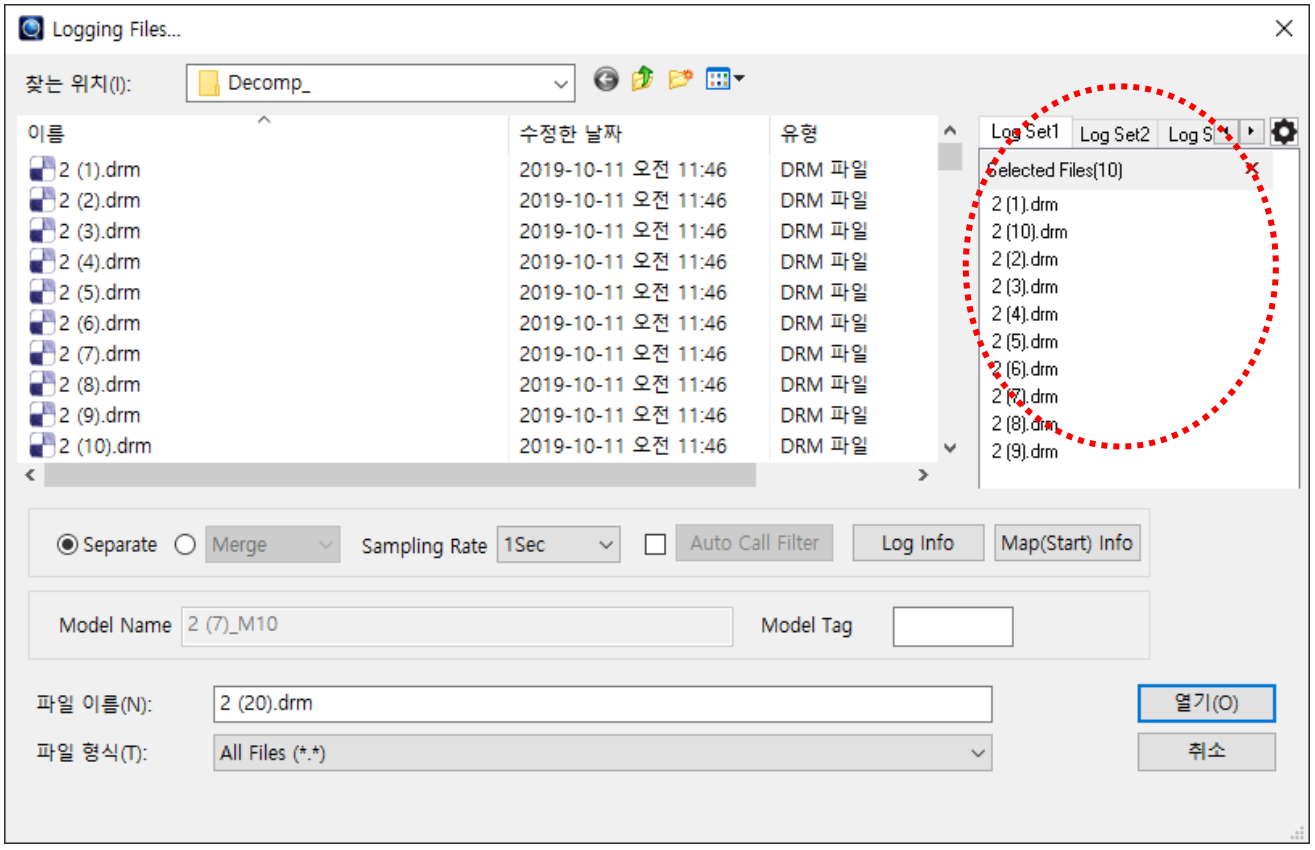
XCAP converts log files collected from DM to a Model which is an analyzable format in various analysis types. XCAP supports various log files formats collected from DM and other measurement tools. Supported log file formats can be found from **Files of Type** in **Logging Files...** window.

Following procedure instructs how to create a new Model (Parsing log data).

1. In XCAP main window, select main Menu bar – **File** – **New**, or click **New** icon in Icon bar.



2. [**Logging Files...**] window appears.

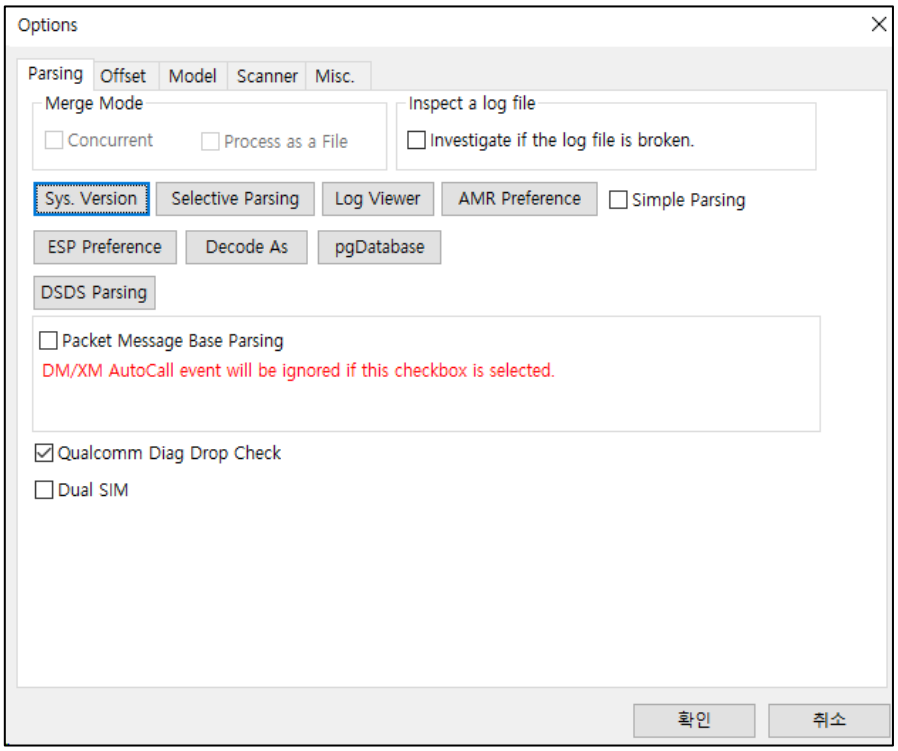


Logging Files Window

3. Drag and drop the log file in File List to **Select Files** field at the right. Multiple log files can be select to create one Model.

- A Model can be created by using several log files in one **Log Set#** tab. Basically 3 **Log Set** tabs are supported as default, and more **Log Set** tabs can be added by clicking **+** tab. Up to 25 **Log Set** tabs can be created. Each Log Set# tab creates one Model folder, and model files can be found in the created folders.
- [**Options**] icon at top right corner enables you to configure further options for creating new models. For more information, see the Chapter 5_Main Window_File_File of the accompanying documentation.

4. To create one model by using several log files, select the radio button for **Separate** or **Merge**.
   - **Merge - Concurrent** option appearing when clicking **Options** icon: Models are created arranging data of several log files in time order.
   - **Separate**: Models can be created for each log files. Parameters are arranged in order of file sequence.
5. **Selective Parsing**: Model can be created using only necessary parameters. (**Options** – **Parsing** tab) For more information, see the Chapter 4_Selective Parsing.pdf of the accompanying documentation.
6. **DSDS Parsing**: Parsing option can be set for Dual SIM Testing. (**Options** – **Parsing** tab) For more information, see the Chapter 4_Usage of DSDS Parsing Option.pdf of the accompanying documentation.
7. **Qualcomm Dual SIM:** Model can be created for Dual SIM Testing. (**Options** – **Parsing** tab) For more information, see the Chapter 5_DSDS Feature.pdf of the accompanying documentation.



8. Configure **Sampling Rate**. Default is set as 1 sec.

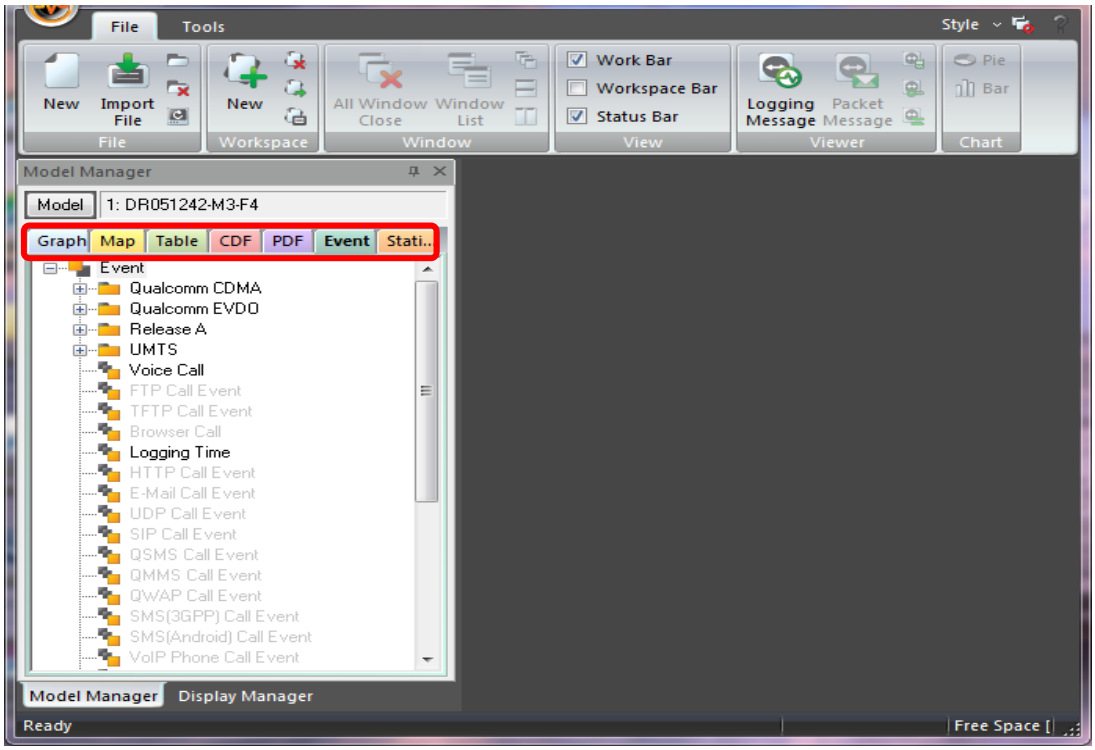
Log data is arranged in accordance with the Sampling Interval. Average of the sampled data is displayed for analysis.

- **All**: Display all collected log data with their respective timestamp information.
- **1 sec**: Lists up timestamp in 1 second unit and each timestamp displays average log data collected for 1 second.
- Sampling Interval is not applied for the parameters for Call Events and Signaling Messages.

9. Enter any information in the entry field of **Model Tag** and the configured Model Tag information is added at the end of model name.
10. Click **Open** button. Model creation is started.
11. Created model file is saved in the program installation folder.

- The storage directory and options can be edited.
- Menu bar – **Tools** – **Options** – **Path** – **Path to load and save a model**

12. When model creation is completed, parameters are activated for each analysis type tabs. XCAP supports 7 analysis types; **Graph**, **Map**, **Table**, **CDF**, **PDF**, **Event**, **Statistics**. In the initial screen, Table analysis tab is opened.



Windows after Model Creation

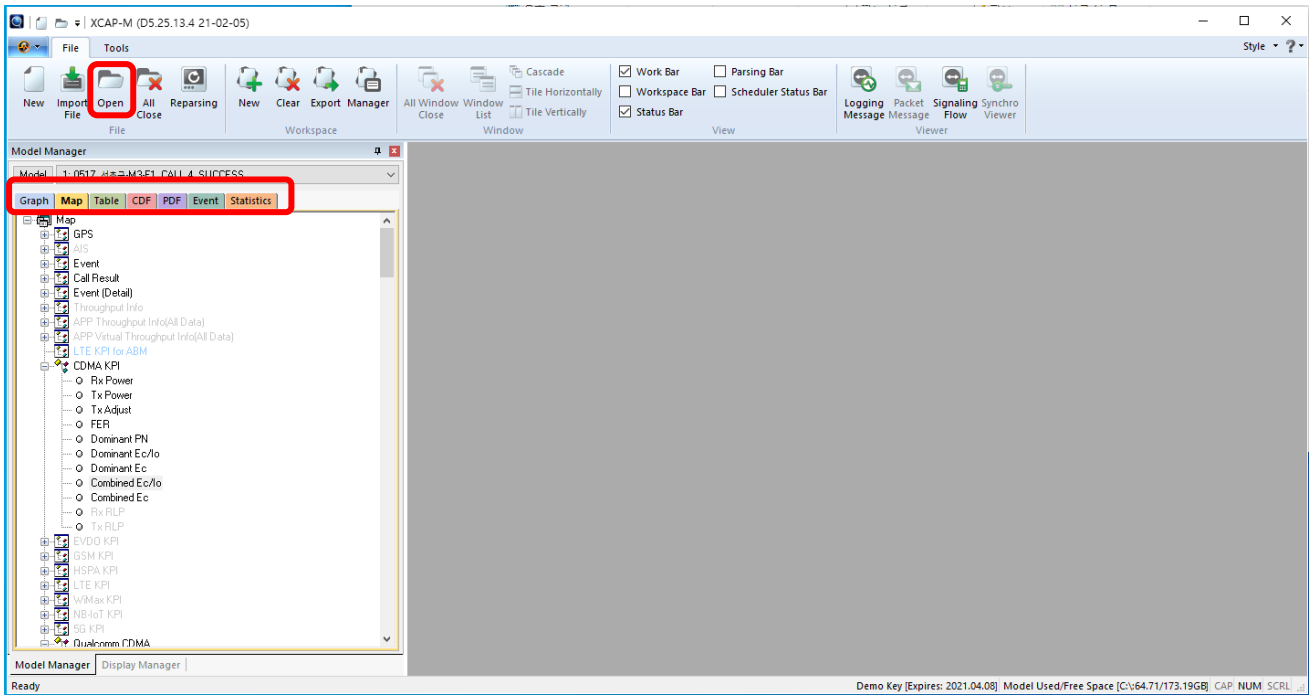


Display Windows of Tabs

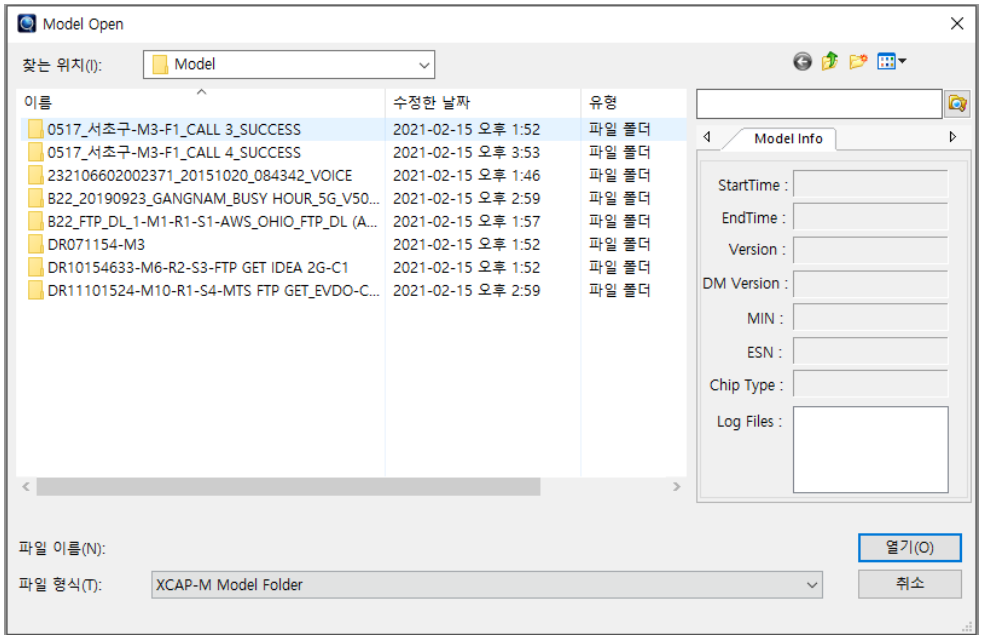
Opening Model

Existing models, which are previously parsed from log data to Model, can be re-opened and analyzed in XCAP. Without parsing log data, existing model can be opened saving parsing time.

1. To open existing Models, select Menu bar – **File** – **Open** or click **Open**  icon in Icon bar.



2. **Model Open** window appears.



3. Select a Model and corresponding Model information is shown in the information field at the

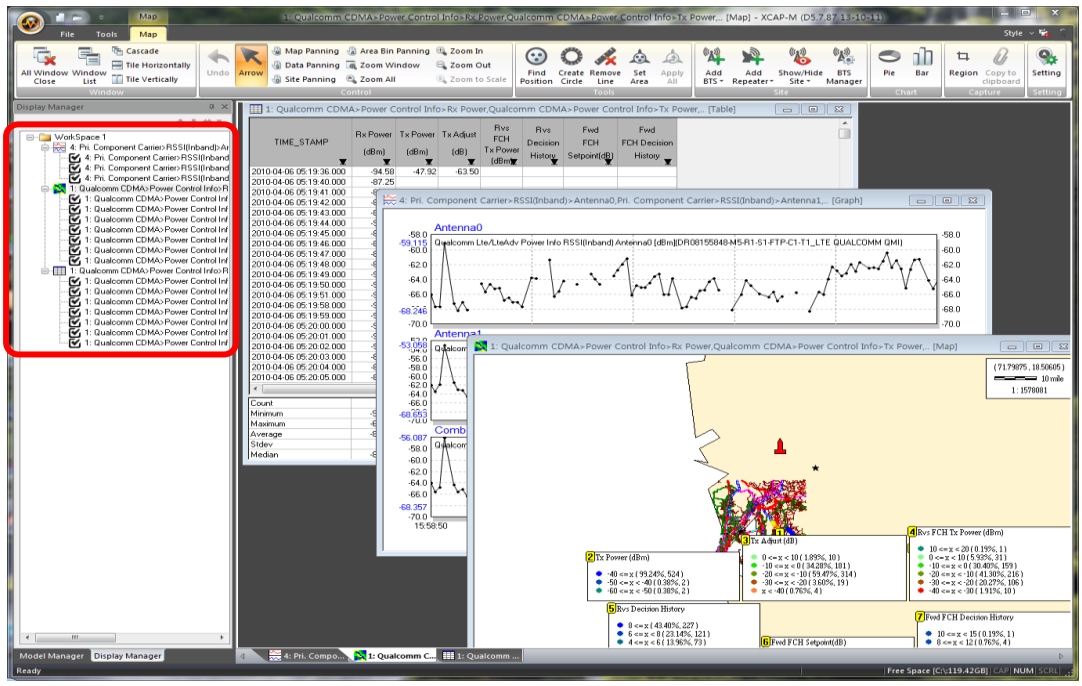
right.

4. Click **Open**.
5. Once a model is created by parsing log data, the created model are displayed in Model Manager at the top right corner. Model button enables you to select other opened model.

Parameter Window Open

Following sub-chapters instruct how to open parameter windows for each analysis type.

1. Parameters are analyzed in various analysis type windows in XCAP. Select an analysis type tab at top left corner.
2. Once an analysis type tab is selected, parameters with data are activated and displayed in black color. Parameters with no data are inactivated displaying in gray color.
3. To open parameter windows, double-click parameters in parameter tree or drag and drop parameters from parameter tree to the workspace at the right.
4. The selected parameters are displayed in **Display Manager** field at the left lower corner sequentially.

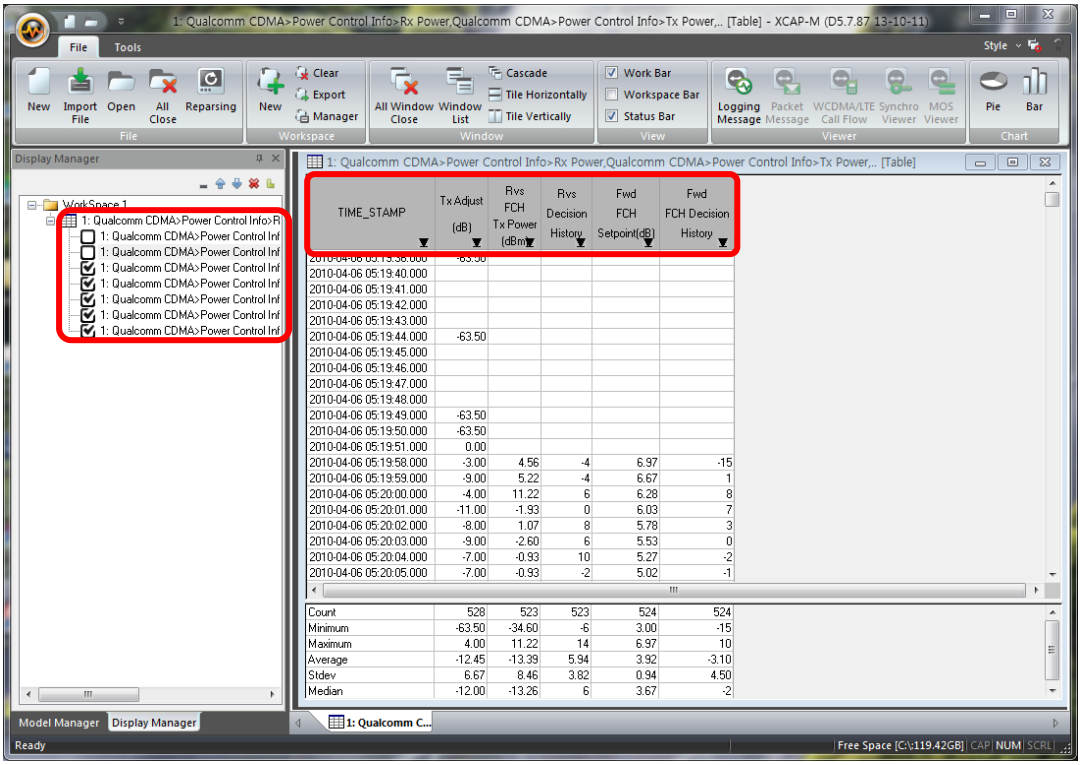


Display Manager of Main Window

5. Parameters selected in **Display Manager** are displayed in the workspace.

- Options for parameter window display can be configured in the following directory.

  Menu bar – **Tools** – **Options**.



Parameter Selection on Display Manager

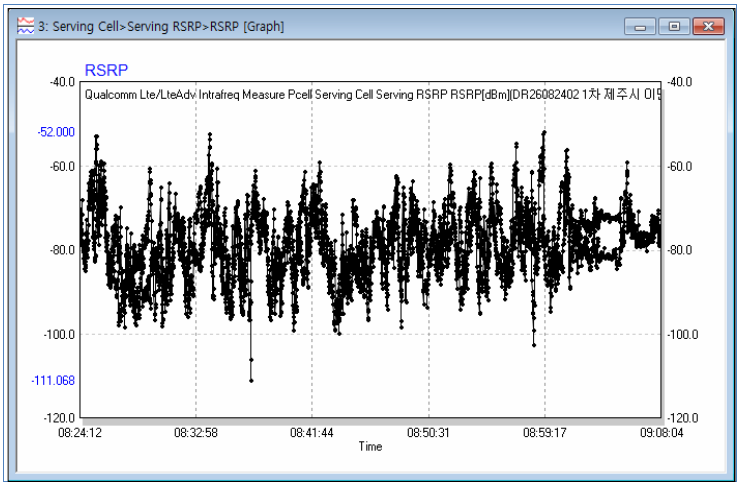
Opening Graph Window

Parameter data are analyzed in Graph window. Following procedure instructs how to open Graph analysis window in the workspace.

1. Select **Graph**  tab at the left.

   - Parameters with analyzable data are activated and displayed in black color. Parameters with no data are inactivated and displayed in gray color.

2. To open parameter windows, double-click parameters in parameter tree or drag and drop parameters from parameter tree to the workspace at the right.



3. X axis shows time and Y axis shows data.

   - Colors, margins, X/Y axis, threshold line etc. in Graph analysis window can be edited in the following directory.

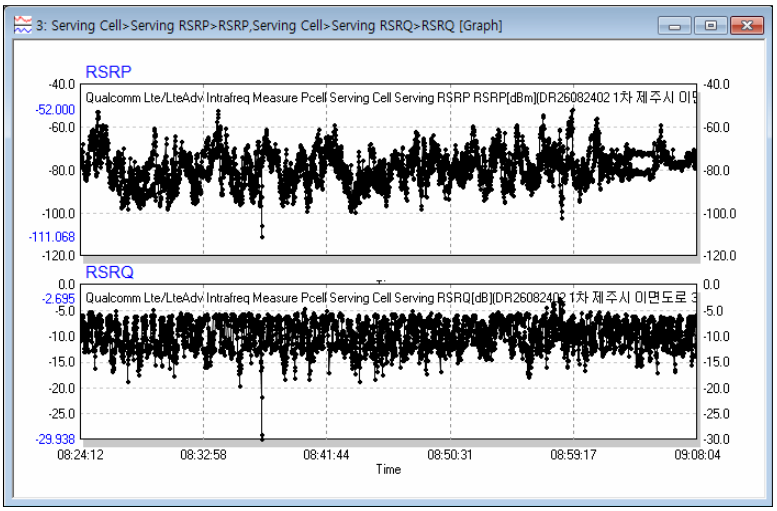
     Menu bar – **Tools** – **Options** – **Graph** tab

4. If dragging and dropping a mother parameter showing with +  from the parameter tree to the workspace, all parameters under the mother parameter are opened in one Graph window in the workspace.

   - To remove a parameter data in Graph window, right-click on the Graph window and select **Remove Graph**.

5. More than two parameters can be displayed in one Graph window. Drag and drop more parameters into the opened Graph window in the workspace.

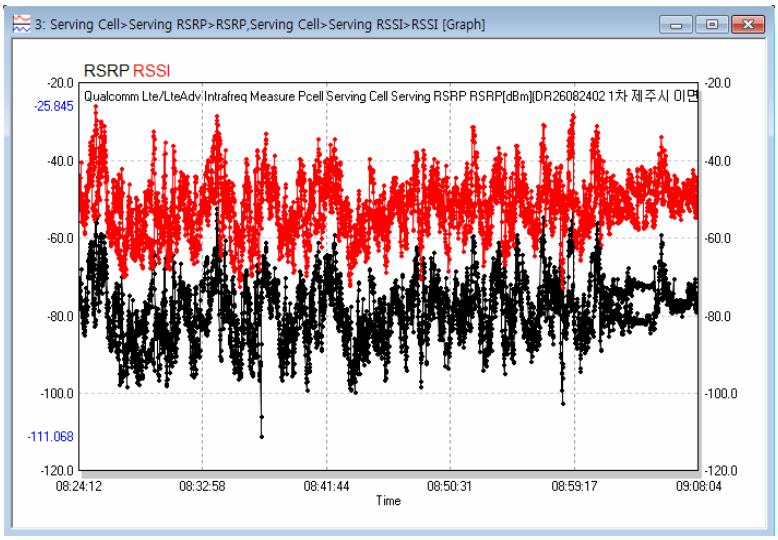
   The added parameter data are placed in the bottom of existing Graph in Graph window.



6. Multiple parameters can be overlaid in one X/Y axis in Graph window. Open a Graph window,

and drag and drop more parameters in the opened Graph window while pressing Alt button in keyboard.

The added parameter data is displayed in different colors in Graph window. Up to 10 parameters can be displayed in one Graph window. It is unable to remove certain parameter in overlaid Graph window.



Opening Map Window

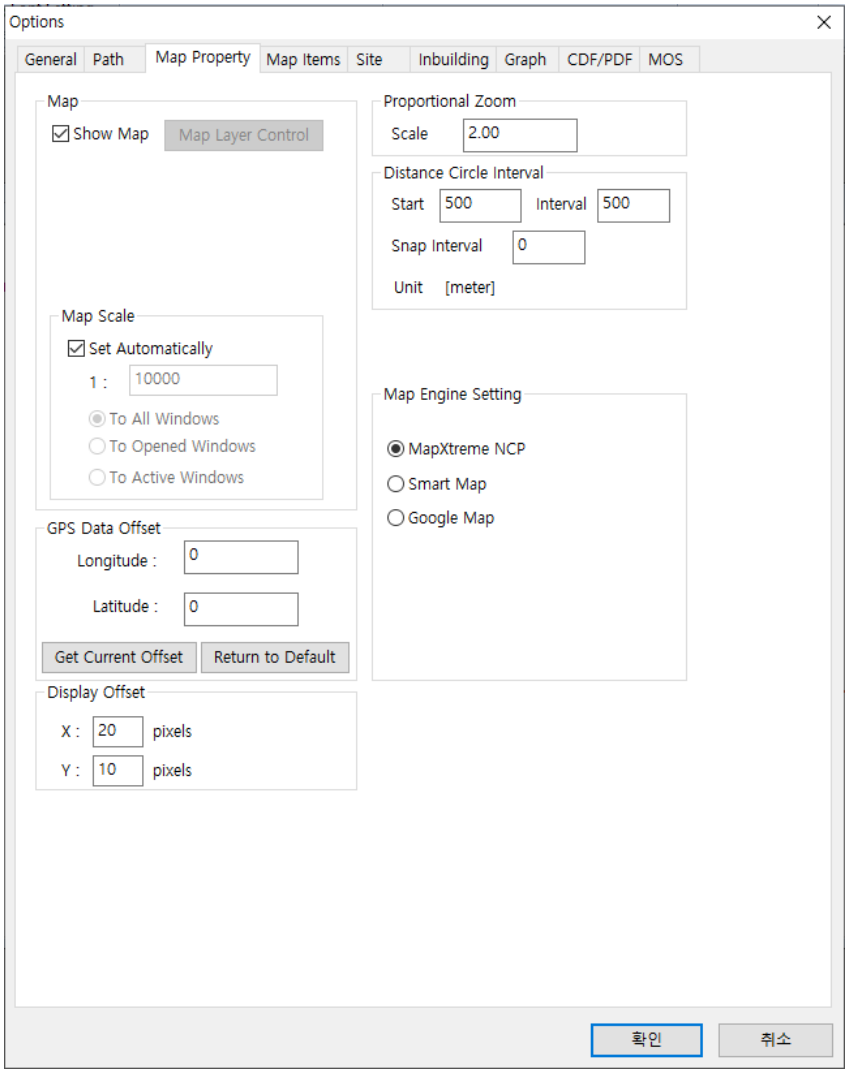
To display and analyze data in map screen, map engine should be installed in the PC and corresponding map engine option should be configured in XCAP in advance.

XCAP supports MapX and MapXtreme engine types.

Following procedure instructs how to open Map analysis window in the workspace.

Configuring Map Engine

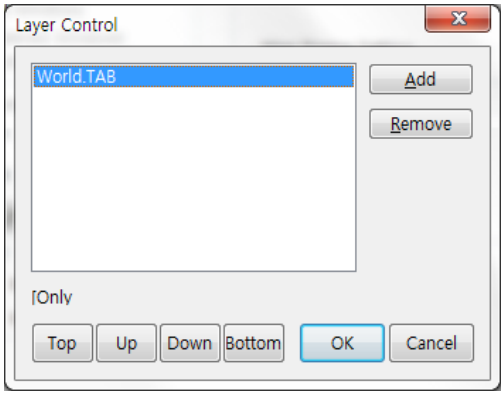
1. Select a Map Engine.
   - Menu bar – **Tools** – **Options** – **Map Property** tab – **Map Engine Setting**.



2. Restart XCAP program to apply map engine configuration.

3. To import Map file, click Menu bar - **Tools** – **Options** – **Map Property** tab – **Map** – **Map Layer Control** button.

4. **Layer Control** screen appears.

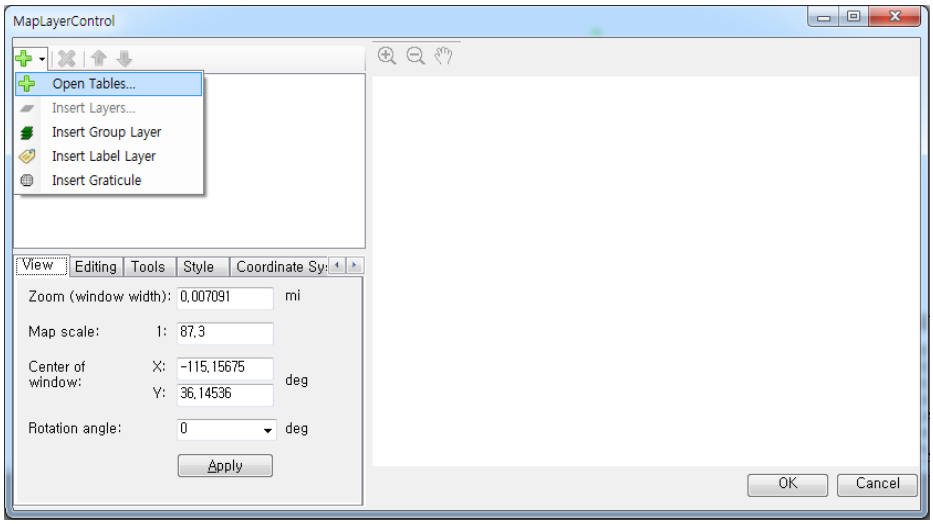


5. Click **Add** button to import corresponding map file.

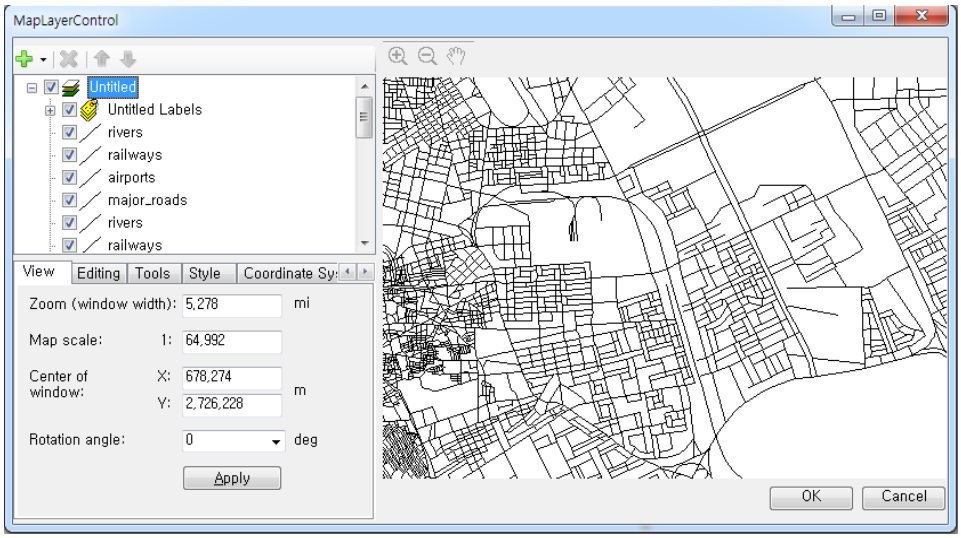
6. Click **OK**.

Configuring MapX Map Engine

1. Open a Map window in the workspace.
2. Right-click in the opened Map window, and select **Configure** – **Map Layer Control**.
3. **Map Layer Control** window appears.



4. Click **Open Table** icon at the left top corner.
5. Select a MapExtreme map file (*.tab) file.



6. Make sure if the selected map file is shown in the Preview screen properly.
7. Click **OK**.

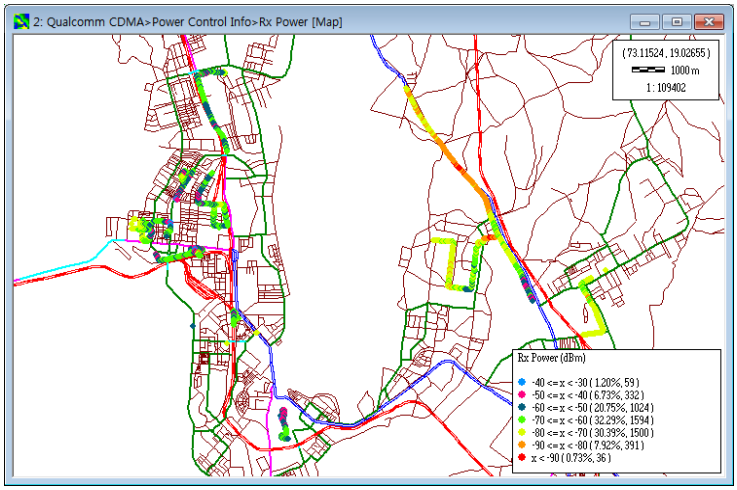
Opening Map Window

Parameter data are analyzed in Map window. Following procedure instructs how to open Map analysis window in the workspace.

1. Select **Map**  tab at the left.

   - Once an analysis type tab is selected, parameters with data are activated and displayed in black color. Parameters with no data are inactivated displaying in gray color.

2. To open parameter windows, double-click parameters in parameter tree or drag and drop parameters from parameter tree to the workspace at the right.



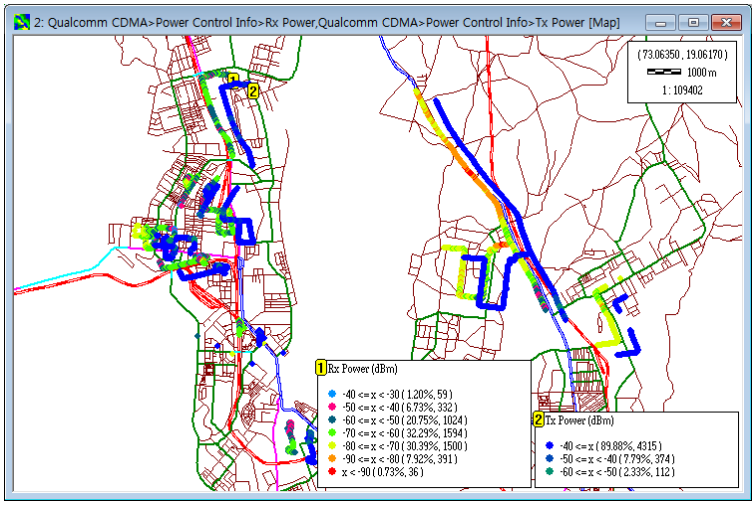
3. Symbols and colors of parameter data displayed in Map window can be edited in the following directory.

   - Map mouse menu – **Configure** – **Legend**
   - Icon bar – **Setting**  icon

4. If dragging and dropping a mother parameter showing with +  from the parameter tree to the workspace, all parameters under the mother parameter are opened in one Map window.

   - To remove a parameter data in Map window, right-click on the Map window and select **Configure** – **Delete Parameter #n**.

5. More than two parameters can be displayed in one Map window. Drag and drop more parameters into the opened Map window in the workspace.



6. The added parameter data are overlapped in Map window. Offset between parameter data in Map window can be configured in the following directory.

   - Menu bar – **Tools** – **Options** – **Map Property** – **Display Offset**

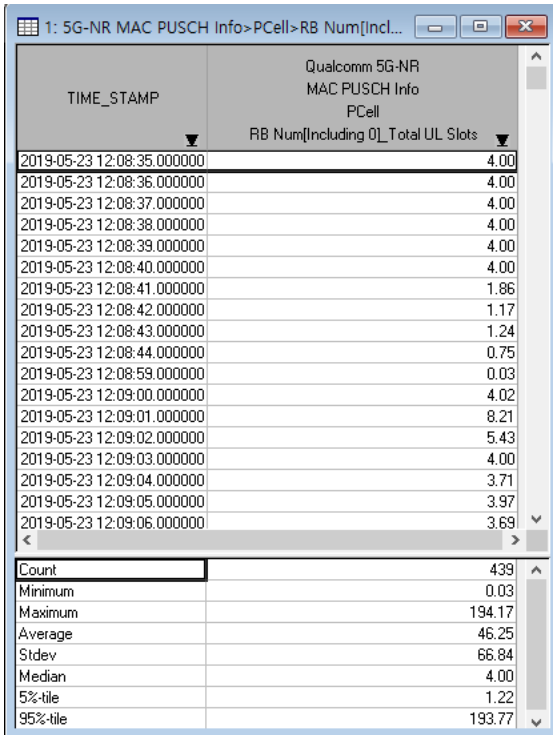
Opening Table Window

Parameter data are analyzed in Table window. Following procedure instructs how to open Table analysis window in the workspace.

1. Select **Table**  tab at the left.

   - Once an analysis type tab is selected, parameters with data are activated and displayed in black color. Parameters with no data are inactivated displaying in gray color.

2. To open parameter windows, double-click parameters in parameter tree or drag and drop parameters from parameter tree to the workspace at the right.



3. Table window includes Data table at the top and Statistics table at the bottom.

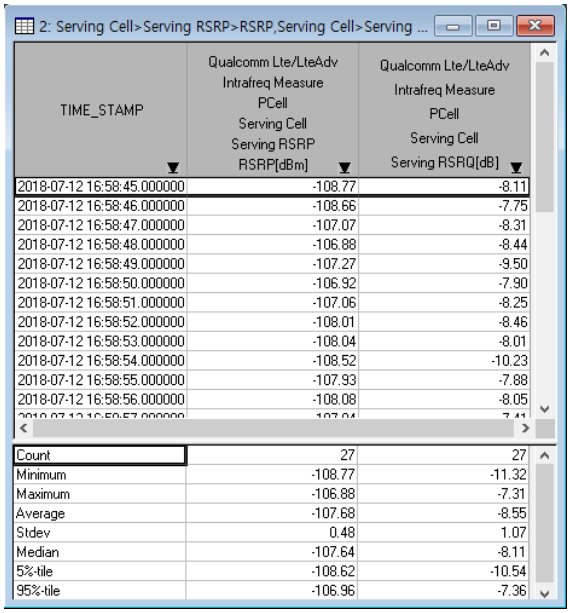
   - Data table: At the top. Basically data are displayed in spreadsheet placing 1-second time unit in X axis vertically. Event names can be organized for X axis in accordance with parameter type.
   - Statistics table: At the bottom. Overall statistical data are displayed including number of data, minimum, maximum, average, standard deviation, and etc. Some parameter may display count statistics in this section.
   - **Show/Hide Statistics** of Table mouse menu: Statistics table can be shown or hidden by using **Show/Hide Statistics** table mouse menu.

4. If dragging and dropping a mother parameter showing with **+**  from the parameter tree to the workspace, all parameters under the mother parameter are opened in one Table window.

   To remove a parameter data in Table window, right-click on the Table window and select **Column Delete**.

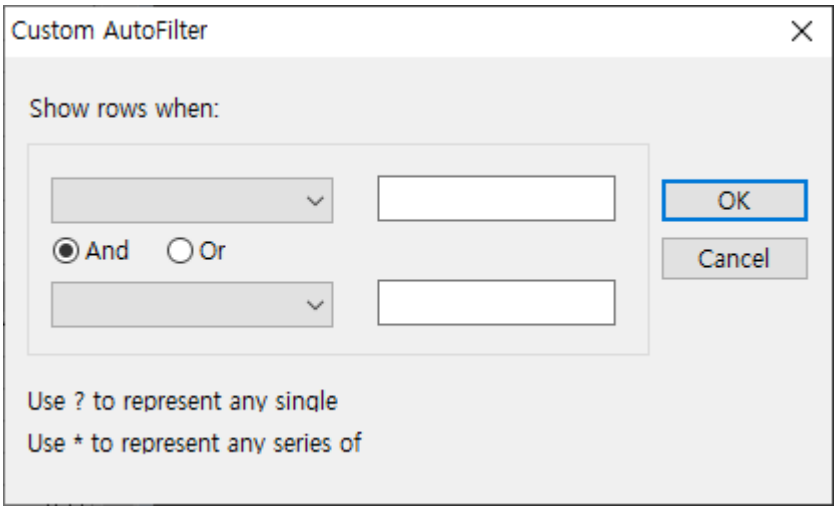
5. More than two parameters can be displayed in one Table window. Drag and drop more parameters into the opened Table window in the workspace.

   The added parameter data are added on the right.



- In multi parameter Table window, the firstly opened parameter column cannot be removed.

6. Table data can be sorted and filtered by using drop-down menu at head columns.
   - **Sort Ascending**: Sort data in ascending order.
   - **Sort Descending**: Sort data in descending order.
   - **All**: Display all data unsorting and unfiltering.
   - **Custom**: Display data in certain condition. Select condition formula (<>, =, Exclude, Include) on the left and enter data in the entry field on the right (character, number).



7. To show the parameter data in PIE and Bar graph window, click PIE or BAR icon in Table Icon bar.

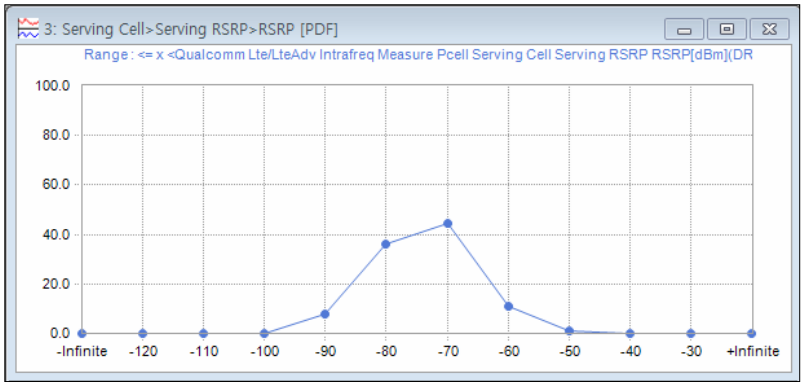
Opening CDF/PDF Graph Window

Parameter data are analyzed in CDF (Cumulative Distribution Function) and PDF (Probability Density Function) graph window. Following procedure instructs how to open CDF/PDF analysis window in the workspace.

As the display options, basic functions, and usages of CDF and PDF graph are the same, CDF/PDF graph are described together in this sub-chapter.

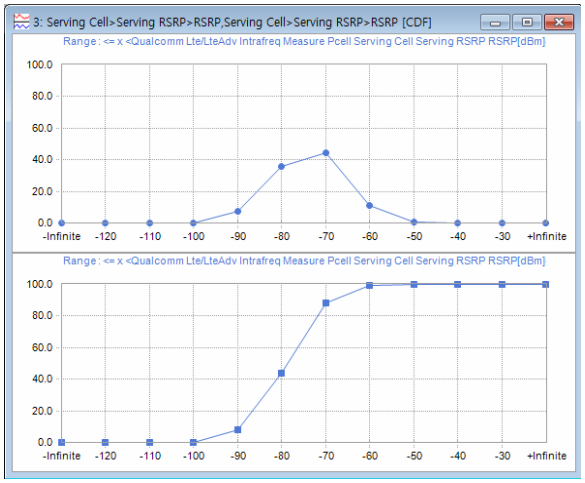
1. Select [**CDF/PDF**] 

tab at the left.
   - Once an analysis type tab is selected, parameters with data are activated and displayed in black color. Parameters with no data are inactivated displaying in gray color.
2. To open parameter windows, double-click parameters in parameter tree or drag and drop parameters from parameter tree to the workspace at the right.



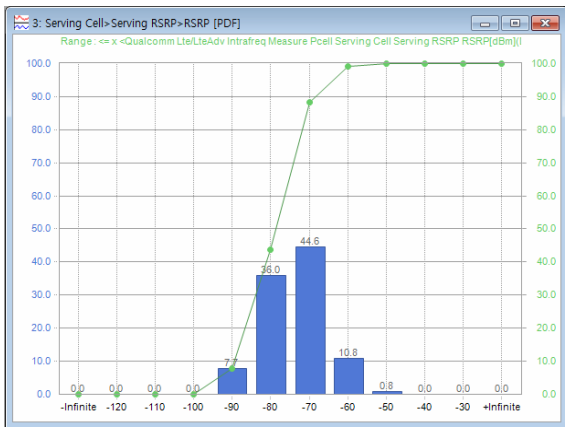
3. X/Y axis in CDF/PDF graph window can be edited in the following directory.
   - CDF/PDF graph mouse menu - **Setting...**
4. If dragging and dropping a mother parameter showing with +  from the parameter tree to the workspace, all parameters under the mother parameter are opened in one Graph window in the workspace.
   - To remove a parameter data in Graph window, right-click on the Graph window and select **Remove Graph**.
5. More than two parameters can be displayed in one Graph window. Drag and drop more parameters into the opened Graph window in the workspace.

   The added parameter data are placed in the bottom of existing Graph in Graph window.



6. Multiple parameters can be overlaid in one X/Y axis in Graph window. Open a Graph window, and drag and drop more parameters in the opened Graph window while pressing Alt button in keyboard.

   The added parameter data is displayed in different colors in Graph window. Up to 10 parameters can be displayed in one Graph window. It is unable to remove certain parameter in overlaid Graph window



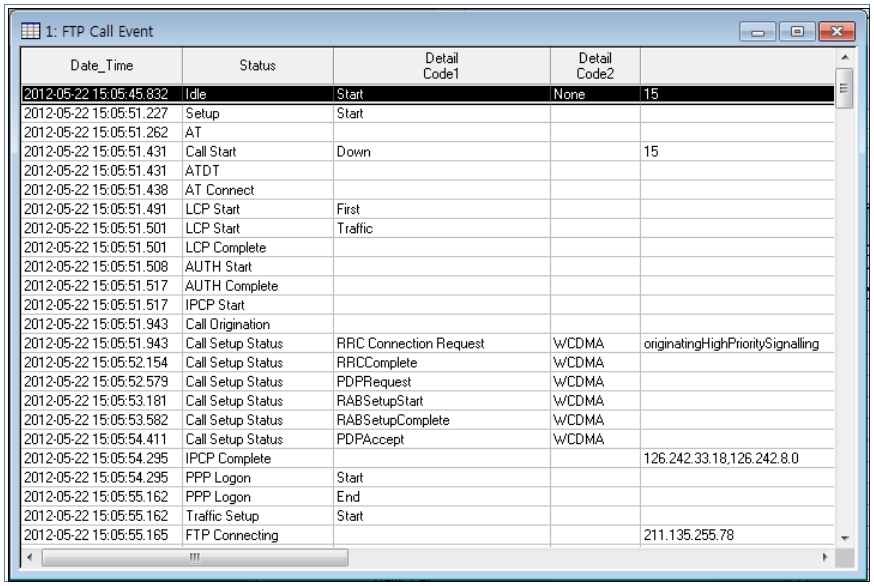
Opening Event Window

Detailed event information in log files is displayed in Event analysis windows. Event analysis window displays all events in log file in list.

1. Select [**Event**]  tab at the left.

   - Once an analysis type tab is selected, parameters with data are activated and displayed in black color. Parameters with no data are inactivated displaying in gray color.

2. To open parameter windows, double-click parameters in parameter tree or drag and drop parameters from parameter tree to the workspace at the right.



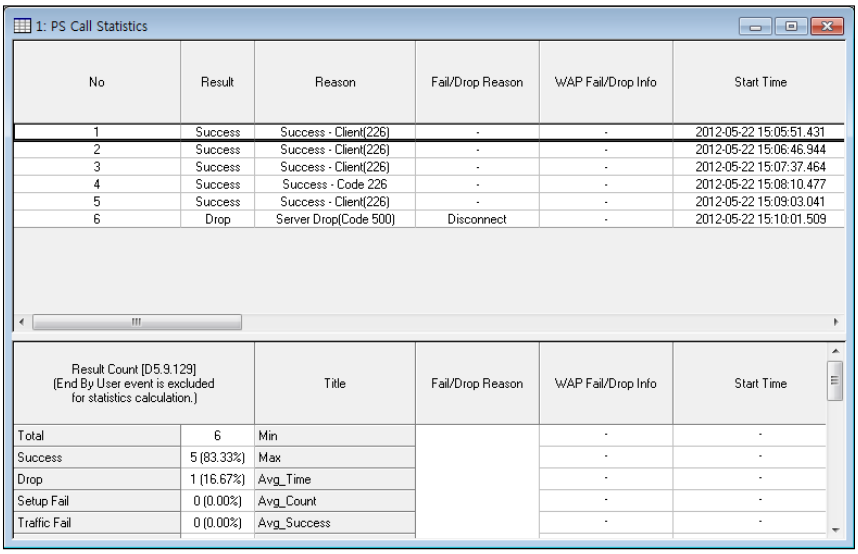
Date_Time	Status	Detail Code1	Detail Code2	
2012-05-22 15:05:45.832	Idle	Start	None	15
2012-05-22 15:05:51.227	Setup	Start		
2012-05-22 15:05:51.262	AT			
2012-05-22 15:05:51.431	Call Start	Down		15
2012-05-22 15:05:51.431	ATDT			
2012-05-22 15:05:51.438	AT Connect			
2012-05-22 15:05:51.491	LCP Start	First		
2012-05-22 15:05:51.501	LCP Start	Traffic		
2012-05-22 15:05:51.501	LCP Complete			
2012-05-22 15:05:51.508	AUTH Start			
2012-05-22 15:05:51.517	AUTH Complete			
2012-05-22 15:05:51.517	IPCP Start			
2012-05-22 15:05:51.943	Call Origination			
2012-05-22 15:05:51.943	Call Setup Status	RRC Connection Request	WCDMA	originatingHighPrioritySignalling
2012-05-22 15:05:52.154	Call Setup Status	RRCComplete	WCDMA	
2012-05-22 15:05:52.579	Call Setup Status	PDPRequest	WCDMA	
2012-05-22 15:05:53.181	Call Setup Status	RABSetupStart	WCDMA	
2012-05-22 15:05:53.582	Call Setup Status	RABSetupComplete	WCDMA	
2012-05-22 15:05:54.411	Call Setup Status	PDPAccept	WCDMA	
2012-05-22 15:05:54.295	IPCP Complete			126.242.33.18,126.242.8.0
2012-05-22 15:05:54.295	PPP Logon	Start		
2012-05-22 15:05:55.162	PPP Logon	End		
2012-05-22 15:05:55.162	Traffic Setup	Start		
2012-05-22 15:05:55.165	FTP Connecting			211.135.255.78

Opening Statistics Window

Detailed statistical information of all calls in log files is displayed in Statistics analysis window. Statistics analysis window displays all statistical data in log file in list.

1. Select [**Statistics**] 

tab at the left.
   - Once an analysis type tab is selected, parameters with data are activated and displayed in black color. Parameters with no data are inactivated displaying in gray color.
2. To open parameter windows, double-click parameters in parameter tree or drag and drop parameters from parameter tree to the workspace at the right.

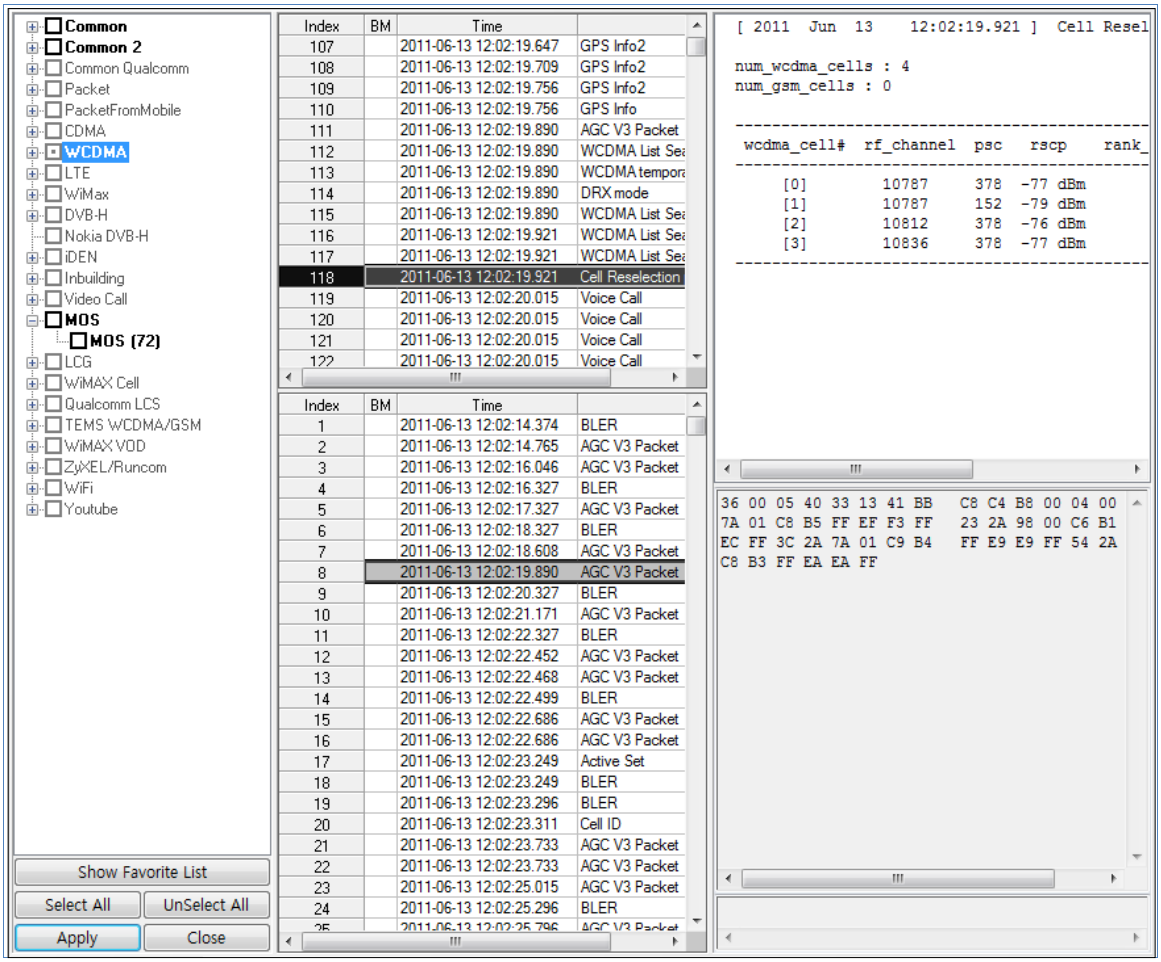


Opening Message Window

Detailed information of all log messages measured through DM is displayed in Message analysis window. Detailed information of selected log message is displayed in list along with corresponding Hex Code and Bit Code for further analysis.

As the display options, basic functions, and usages of Logging Message and Packet Message windows are the same, Logging/Packet message windows are described together in this sub-chapter.

1. Select File – Logging/Packet Message
2. Select main Menu bar – **Message** – **Logging/Packet Message**, or click **Logging/Packet Message**  icon in main Icon bar.



Field	Location	Description
Filter Tree	At the left	The selected parameters in the filter tree are displayed in Filtered Message View field. Select desired parameters from the Fitler Tree and click Apply button.
All Message View	At the top center	All log messages are displayed in list.
Filtered Message View	At the lower center	Filtered log messages are displayed in list.
Raw Data Parsing information View	At the top right	Raw data of the messages selected in Filtered Message View is displayed.
Hex Code	At the right center	Corresponding Hex Code information of the raw data is displayed.
Bit	At the lower right	Corresponding Bit information of Hex Code is displayed.

3. Top center field displays all log message information.
4. Select a parameter in the filter tree at the left, and click **Apply** button at the bottom.

   All messages of the selected parameters are listed up in Filtered Message View filed in the lower center.

   - Filter Tree may vary depending on Model. **Close** button the at bottom hides filter tree.

5. Double-click a log message in the Filtered Message View field in the lower center, and corresponding Raw Data Parsing information is displayed in the top right.
6. Hex Code displayed at the right center can be used for further analysis.

   If dragging and dropping an Octet or double-clicking an Octet in Hex Code field at the right center, corresponding Bit information is displayed at the lower right field. Bit of Hex Code can be used for further analysis.

Analysis Method

XCAP analyzes various data in various analysis formats using multiple analysis methodologies.

This chapter explains commonly used analysis methodologies of XCAP Analysis.

- **Model Creation Plan (Scheduler)**
- **Message Duration Time Analysis**
- **LTE RB Scheduling Map Usage**
- **Compare Before and After Delta Map Optimization**
- **M to M Statistics Creation**
- **Filtering**
- **Selective Parsing**
- **Editing GPS Coordinate**
- **Create Virtual GPS Function**
- **Exporting Batch using Batch Export Feature**
- **Area Binning**
- **PSYNC Pollution Analysis in Qualcomm Tree**
- **PSC Pollution Analysis**
- **Neighbor List Analysis**
- **Finding Missing Neighbors**
- **Serving Cell Line Drawing**

© Innowireless Co., Ltd.

- **Auto Reporting with Map Legend Change**
- **Playing MOS Wave File**
- **Voice Wave Silence Detection**
- **Map Display of Inbuilding Data**
- **Event Call Extract from Statistics Window**
- **Add User Column Usage**
- **Usage of Favorite Info**
- **File Export using Command Prompt**
- **Multi Process**
- **User Defined Report**
- **LTE DCI Analysis**
- **User Defined KPI Parameter**
- **pgDatabase Usage**

For more information, refer to each of the Analysis Method documents that come with it.

Basic Menu

This chapter describes Basic Menus of XCAP, to analyze various tabs, trees and windows.

There are three basic menus in here:

- Main Window
- Window Mouse Menu
- Tree Mouse Menu

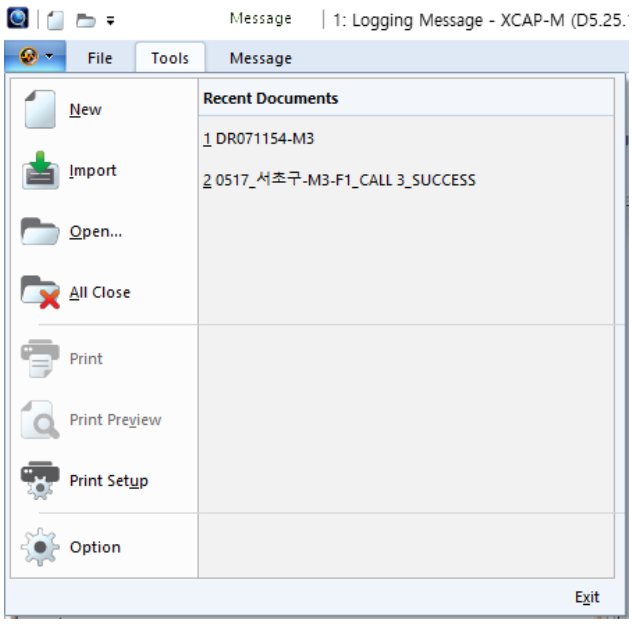
Main Window

Main window of XCAP is composed of five sectors;

- Main Menu: Enable to use basic menu for XCAP.
- File: Enable to use open/show/hide parameter window.
- Tools: Enable to use major function with icons for analysis.
- Window Bar: Enable you to use major functions with icons. Icons vary depending on analysis type; Graph and Map.
- Work Bar: Show model name and parameter trees for each analysis type. And allow you to analyze parameters in the space.

Main Menu

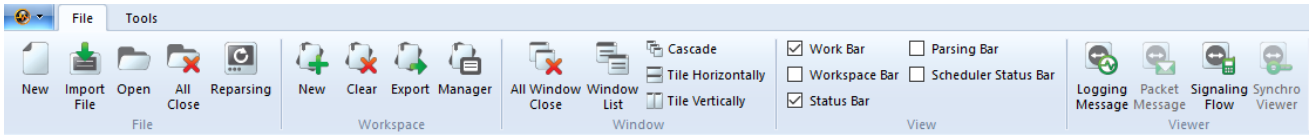
When you select  button, the following menus are available.



Field	Description
New	Create a new model using a logging file.
Import	Import saved file.
Open	Display existing model.
All close	Close all models opened.
Print	Print work space.
Print Preview	Preview printed view.
Print Setup	Configure for printing jobs.
Option	Set up basic options for each analysis type.
Recent Documents	Display recent opened document.

File

Following six function groups are available with File menu.



File Menu Bar

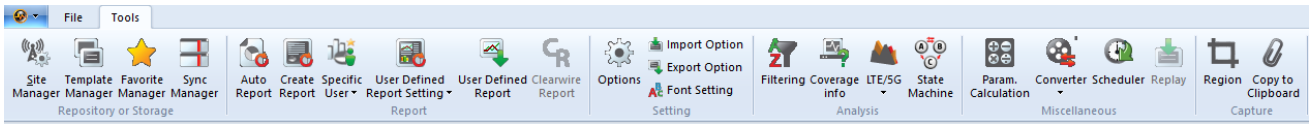
The following table is information of File menu.

Classification		Description
File	Create Model, import files, open model, model all close and reparsing log. For more information, see Chapter 5_Main Window_File_File.pdf. that comes with it.	
	New	Create a new model using a logging file.
	Import File	Import data file (*.csv or *.txt) to analyze. Imported data is

		shown under Import Data parameter tree for each analysis type.
	Open	Display existing model.
	All Close	Close all models opened.
	Reparsing	Reparse messages you select, and create a new model. Reparsing function is available only when particular messages are selected from Selective Parsing window during new model creation.
Workspace	New	Open new workspace as tab.
	Clear	Close all workspaces.
	Export	Export current parameter windows in workspace into Excel or PowerPoint.
	Manager	Manager workspace list.
Window	All Window Close	Close all parameter windows opened in workspace.
	Window List	Show a list of currently opened parameter windows.
	Cascade	Arrange parameter windows in workspace tiered. (Shortcut: F10)
	Tile Horizontally	Arrange parameter windows in workspace horizontally. (Shortcut: F11)
	Tile Vertically	Arrange parameter windows in workspace vertically. (Shortcut: F12)
View	Work Bar	Show/hide Model Manager and Display Manager.
	Workspace Bar	Show/Hide Workspace.
	Status Bar	Show/hide Status bar. Status bar shows windows' working status at the bottom of main window.
	Parsing Bar	Show parsing status
	Scheduler Status bar	Show scheduler status
Viewer	Open message viewers as below. For more information, see Chapter 5_Main Window_File_Viewer.pdf that comes with it.	
	Logging Message	Open Logging Message window. (Shortcut: Ctrl + L)
	Packet Message	Open Packet Message window showing packet data (Modem/Network adapter) logged from DM. (Shortcut: Ctrl + Q)
	Signaling Flow	Check Call Flow for Layer3 Signaling Message. It contains direction between messages, Piggybacked NAS data and Itra LTE HO data. (Support LTE, WCDMA, GSM)
	Synchro Viewer	Show Synchro Viewer (TCP Flow) of logging file. Activated only when Modem/Network adapter data is included in Logging file. Note) Modem/Network adapter data can be parsed by selecting Selective Parsing in Logging Files… window.
Chart	Pie	Display Pie chart for current window.
	Bar	Display Bar chart for current window.

Tools

Following six function groups are available with Tools menu.



Tools Menu Bar

The following table is information of File menu.

Classification		Description
Repository or Storage	Site Manager	Add, delete, and edit BTS and Repeater. For more information, see Chapter 5_Main Window_Tools_Site Manager.pdf that comes with it.
	Template Manager	Manage template supporting each pop-up menu of graph, table, etc.
	Favorite Manager	Manage the Favorite items created by Table, Graph and Map.
	Sync Manager	Synchronized data per call to analyze with Graph, Map, Table. For more information, see Chapter 5_Main Window_Menu Bar_Tools_Sync Manager.pdf that comes with it
Report	Create various reports as below. For more information, see Chapter 5_Main Window_Tools_Report.pdf that comes with it.	
	Auto Report	Start Auto Report.
	Create Report	Create report in pdf, excel, doc, html, and etc.
	Specific User	Create report for particular users
	User Defined Report Setting	Design report files.
	User Defined Report	Publish report files.
Setting	Configure options. For more information, see Chapter 5_Main Window_Tools_Setting_Options.pdf that comes with it.	
	Options	Set up basic options for each analysis type.
	Import Option	Apply settings saved as "*.oxf".
	Export Option	Export settings for Filter List, Favorite, Correlation, Auto Report, Legend, Options, for each analysis, and etc. to "*.oxf" file. The exported "*.oxf" file may be imported again.
	Font Setting	Configure the user desired font style and size.
Analysis	Supports analysis method. For more information, see the Chapter 5_Main Window_Tools_Analysis.pdf that comes with it.	
	Filtering	Filter data in accordance with user-selected-options.
	Coverage Info	Analyzes data quality (Good, Bad, No Service, etc.) based on RSSI, Ec/Io parameter, and shows on map analysis windows.

	LTE/5G	◆ LTE RB scheduling Map: LTE Resource Block Scheduling. ◆ LTE DCI Analysis: Analyzes detailed data per TTI ◆ LTE PCI Match Ratio: When LTE Carrier Aggregation Cell planning and optimizing, the same PCI is determined for each frequency and the same Ratio is shown on the table and map. ◆ LTE Shadow Area Estimation: In the LTE log, the threshold value of the RSRP/RSRQ/RSSI/SNR is used to determine the shaded region if the signal strength is below the reference value. ◆ LTE Cell Throughput: Analyzes cell throughput data while conducting small cell test by using XCAL-MTS. ◆ 5G-NR Cell Throughput: It is a function to calculate the cell throughput by summing the throughput of multiple terminals measured at the same time ◆ 5G-NR DSS: It is a function to calculate the sharing ratio within the same frequency band for LTE and 5G. ◆ Sector Swap Analysis: Identify cross-sector intrusion and Sector Swap information. ◆ 5G-NR Beamforming information: Serving Beam and Non-Serving Beam information are provided in two dimensions through the Tx Beam SSB Index Array. ◆ 5G-NR SCS Frame Format: In case of 5G-NR TDD, UL/DL Frame Format per slot is shown by referring to tdd-UL-DL-Configuration information.
Miscellaneous	Support additional features. For more information, see Chapter 5_Main Window_Tools_Misc.pdf that comes with it.	
	Param. Calculation	Compute several parameters in Analyzer simply.
	Converter	◆ DRX to DRM, SCX to SCN: Convert DRX to DRM and SCX to SCN. ◆ Convert to DLF (Qc format): Convert log data to DLF using log code defined by ICD after deleting XCAL(Accuver DM Program)'s own Event. (If analyze log using QxDM or QCAP, use the corresponding Converter.) ◆ Convert to PCAP (Ethernet PPP): Extract PCAP data from Log data. (If analyze in Wireshark, use the corresponding Converter.) ◆ Convert to PCAP (Mobile IP Packet): Extract Qualcomm / Samsung chipset PPP Frame log code among total log data. ◆ Convert to DAT (Intel format): This is a function that allows you to change drm file to Anritsu Scanner format (*.dat). It is also possible to change the Anritsu Scanner format (*.dat) to drm. ◆ Convert to ISTP (Intel format): Convert to ISTP format(Intel format). ◆ Convert ISTP to DRM: Convert ISTP format to DRM format. ◆ IntegrationConverter: IntegrationConverter that allows log data to be converted to drx, scx, aox, aoc, etc. or converted to drm files.
	Scheduler	Makes a plan to create Models by using log files in certain

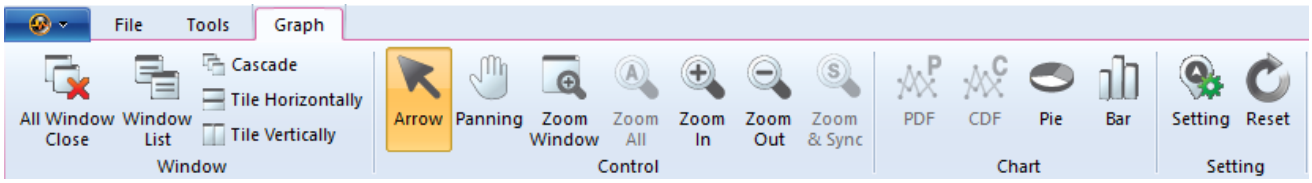
		conditions from pre-designated directory at planned time. For more information, see Chapter 4_Model Creation Plan (Scheduler).pdf that comes with it.
	Replay	Replay data as time order.
Capture	Region	Capture image as region.
	Copy to clipboard	Copy current parameter windows opened.

Window Bar

Enable you to use major functions with icons depending on window types: Graph, Map, Table, CDF, PDF, Event, and Statistics.

Graph Bar

Graph bar includes icons for major functions of graph analysis. Graph bar appears during Graph, CDF, and PDF analysis.



Graph Bar

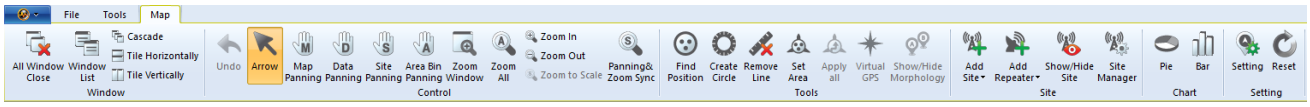
The detailed description of each icon in Graph bar is as follows.

Classification		Description
Window	All Window Close	Close all parameter windows opened in workspace.
	Window List	Show a list of currently opened parameter windows.
	Cascade	Arrange parameter windows in workspace tiered. (Shortcut: F10)
	Tile Horizontally	Arrange parameter windows in workspace horizontally. (Shortcut: F11)
	Tile Vertically	Arrange parameter windows in workspace vertically. (Shortcut: F12)
Control	Arrow	Select data, and move in the analysis window.
	Panning	Move a graph in any direction.
	Zoom Window	Zoom in selected range on a graph.
	Zoom All	Return to default graph scale after zoom in/out.
	Zoom Previous	Go back to previous graph scale.
	Zoom & Sync	Zoom in on selected range with the mouse and synchronize the same data range in different analysis windows.
	Zoon In	Zoom in specific point on a graph.
	Zoom Out	Zoom out specific point on a graph.

Chart	PDF	Show PDF graph for the parameter.
	CDF	Show CDF graph for the parameter.
	Pie	Convert current graph data to Pie graph. (Shortcut: Ctrl + E)
	Bar	Convert current graph data to Bar graph. (Shortcut: Ctrl + B)
Setting	Set up graph, pie, bar graph legend.	
Reset	Returns to the initial template settings. Only active if you use the “Template Manager – Attributes” option.	

Map Bar

Map bar includes icons for major functions of Map analysis.



Map Bar

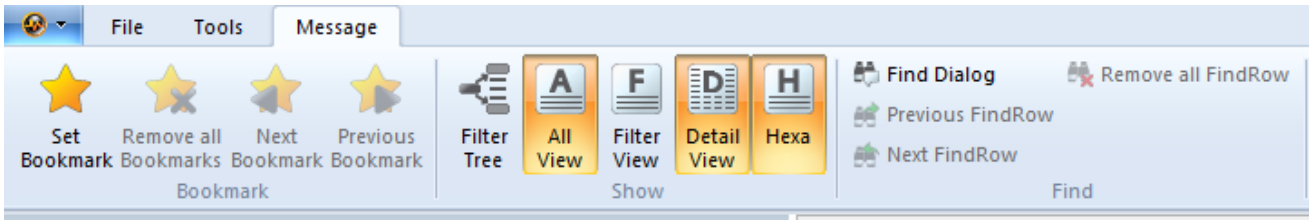
The detailed description of each icon in Map bar is as follows.

Classification		Description
Window	All Window Close	Close all parameter windows opened in workspace.
	Window List	Show a list of currently opened parameter windows.
	Cascade	Arrange parameter windows in workspace tiered. (Shortcut: F10)
	Tile Horizontally	Arrange parameter windows in workspace horizontally. (Shortcut: F11)
	Tile Vertically	Arrange parameter windows in workspace vertically. (Shortcut: F12)
Control	Undo	Undo panning and zooming operation
	Arrow	Select data, and move in the analysis window.
	Map Panning	Move map in any direction.
	Data Panning	Move data in any direction.
	Site Panning	Move Site/Repeater in any direction.
	Area Bin Panning	Enables you to divide sections in designated area, called bin, and extract representative values in each bin.
	Zoom Window	Zoom in specific area on a graph.
	Zoom All	Return to default scale after zoom in/out.
	Zoom In	Zoom in specific point on a graph.
	Zoom Out	Zoom Out: Zoom out specific point on a graph.
	Zoom to Scale	Zoom to scale: Return to default graph scale after zoom in/out.
	Panning&Zoom Sync	It is a function that applies Map Panning, Zoom In/Out effects to multiple map windows. If you display multiple map windows, apply Map Panning, Zoom In/Out in one window, and double-click (Sync) the data with this option clicked, the effect is also applied to the other map windows.
Tools	Find Position	Search a specific point using longitude, latitude, Site/Repeater/Sector/Bin name.
	Create Circle	Measure distance between two points using circle.
	Remove Line	Remove the following lines; Information Tool Tip window, Coverage line, Serving Cell line, Sync line, Find line, Line connecting between mother sector and repeater, Line connecting between neighbor sectors.
	Set Area	Save map setting for a particular area such as zoom and panning.
	Apply All	Apply map setting saved by Set area function to all map

		parameter window.
	Virtual GPS	Adds virtual GPS data for the log files with no GPS information
Site	Add Site	Add Site on current map parameter window for each technology.
	Add Repeater	Add Repeater on current map parameter window for each technology.
	Show/Hide Site	Show/hide Site and Repeater for each technology.
	Site Manager	Manage Site/Repeater, it allows you to add, delete, and edit BTS and Repeater.
Chart	Pie	Convert current graph data to Pie graph. (Shortcut: Ctrl + E)
	Bar	Convert current graph data to Bar graph. (Shortcut: Ctrl + B)
Setting	Set up map legend.	
Reset	Returns to the initial template settings. Only active if you use the “Template Manager – Attributes” option.	

Message Bar

Message bar includes icons for major functions of Map analysis.



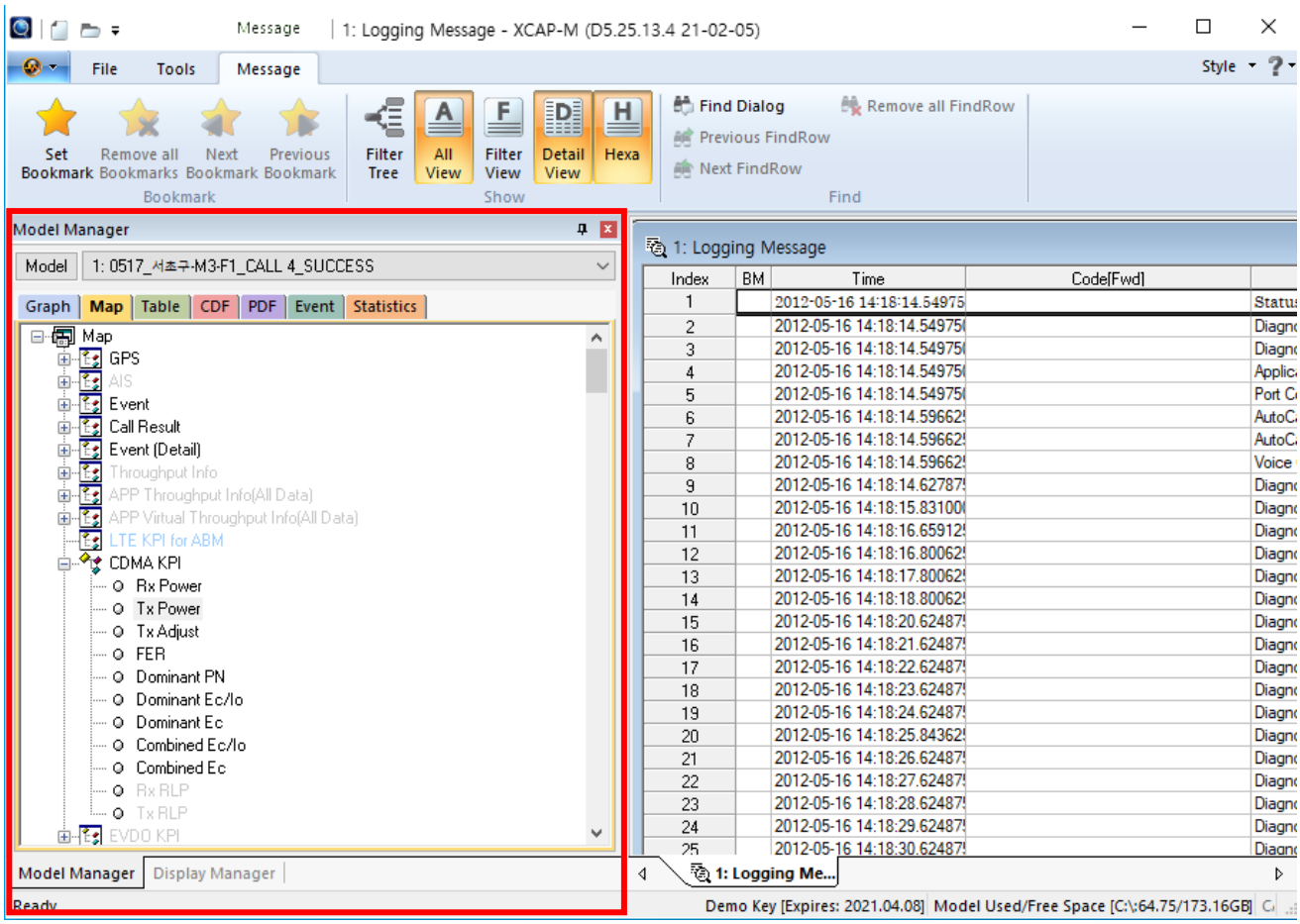
Message Bar

The detailed description of each icon in Map bar is as follows.

Classification		Description
Bookmark	Set Bookmark	Set a bookmark for current message.
	Remove all Bookmarks	Remove all bookmarks set.
	Next Bookmark	Move to next bookmark set.
	Previous Bookmark	Move to previous bookmark set.
Show	Filter Tree	Show filter tree that you may select on the left side of logging message window.
	All View	Show/hide logging message list. All message windows are set to display when you open a logging message window at first.
	Filter View	Show/hide filtered message window.
	Detail View	Show/hide detailed information of message windows selected from All Message View or Filtered Message View window.
	Hexa	Show/hide Hex. Code in Detail Message View window.
Find	Find Dialog	Find dialog
	Previous FindRow	Move to previous string found.
	Next FindRow	Move to next string found.
	Remove all FindRow	Remove all strings found.

Work Bar

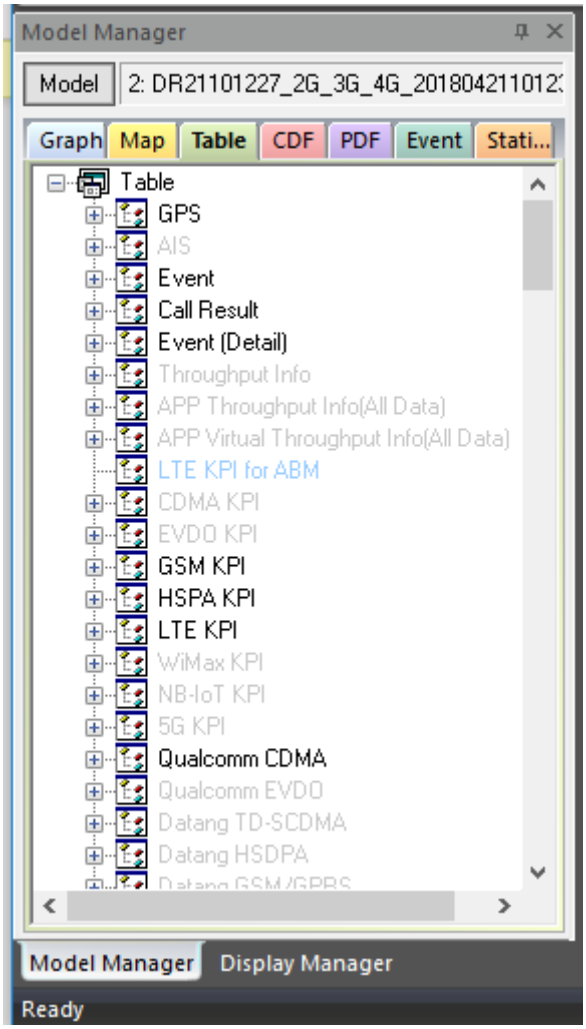
Work bar is divided into two managers, Model Manager and Display Manager, with tab as below.



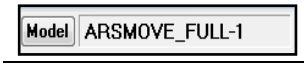
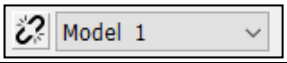
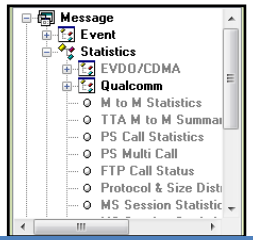
Work Bar

Model Manager

Model Manager enables you to see model name, analysis type, and parameter tree.



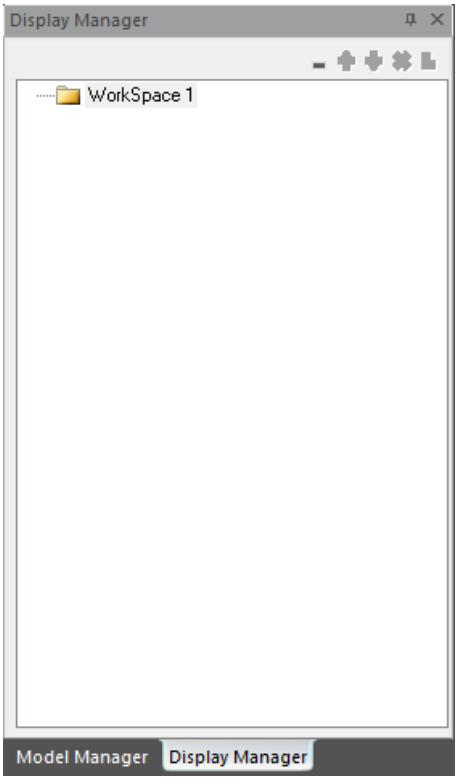
Model Manager

Icon	Name	Description
	Model	Show name of current model opened. Model button allows you to select another model available.
	Model link	The Model link icon allows you to select whether to link or unlink the workspace and model.
	Analysis tab	Include seven analysis types; Graph, Map, Table, CDF, PDF, Event, and Statistics.
	Parameter tree	Show classified parameters as well as organized parameter trees. Parameter tree may vary depending on model selected.

Display Manager

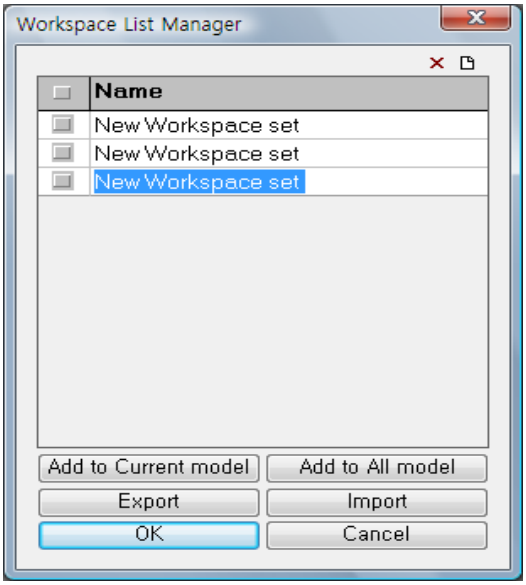
Display Manager is a helper that allows you to manage workspace quickly and easily. It enables you to move another model and manage workspace.

Workspace bar is viewable by clicking Display Manager tab of main window.



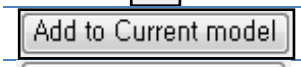
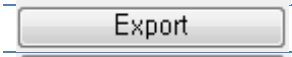
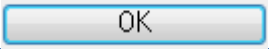
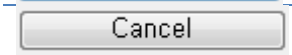
Workspace Manager

By clicking  Workspace Manager icon, workspace list manager appears.



Workspace List Manager

The detailed description of each icon in Workspace List Manager is as follows.

Icon	Name	Description
	Delete	Delete the selected workspace.
	New	Create a new workspace.
	Add to Current model	Add selected workspace to current model opened.
	Add to All model	Add selected workspace to all model opened.
	Export	Export selected workspaces.
	Import	Import workspace in wrk format.
	OK	Apply changed settings.
	Cancel	Cancel changed settings and close the window.

Note) Current job settings in a workspace such as window size, position, and etc. is automatically saved when you move to another workspace tab. It also applies when you create a new workspace.
When you import another model's workspace, current workspace is automatically saved and a new workspace is opened importing data.
It is impossible to import workspaces for a deleted model.

Window Mouse Menu

There are 6 tabs in XCAP and each of windows has their own mouse. In addition Logging Message window has either. Therefore herein describes of them.

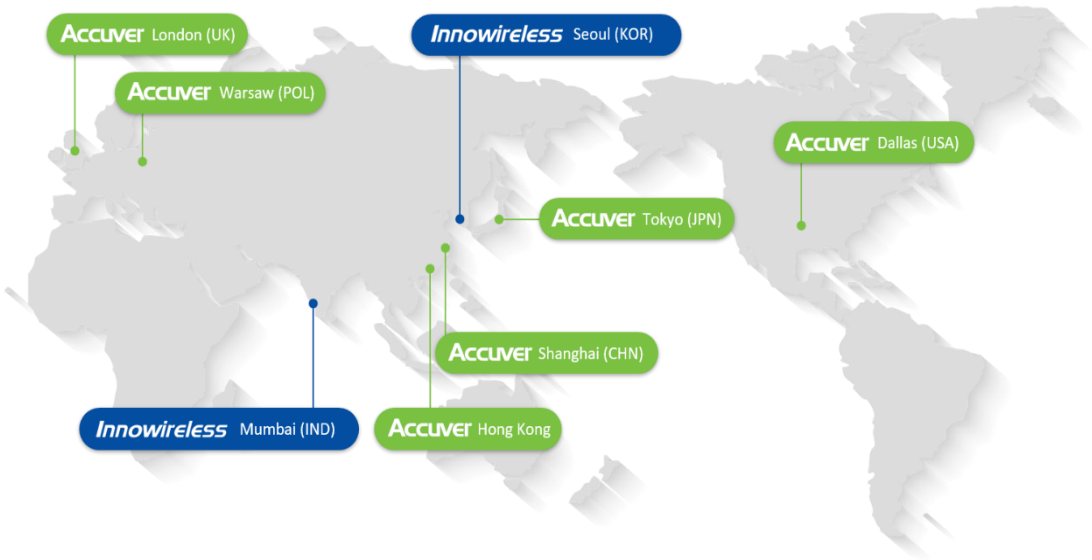
Classification	Description
Graph	Change the properties of Graph or use the various functions for Parameter Graph Window. For more information, see Chapter 5_Mouse Menu_Graph.pdf that comes with it.
Table	Change the properties of table or use the various functions for Parameter Table Window. For more information, see Chapter 5_Mouse Menu_Table.pdf that comes with it.
Map	Change the properties of table or use the various functions for Parameter Map Window. For more information, see Chapter 5_Mouse Menu_Map.pdf that comes with it.
Message	Change the properties of table or use the various functions for Logging Message Window. For more information, see Chapter 5_Message Menu_Map.pdf that comes with it.
CDF/PDF	Change the properties of CDF/CDF Graph or use the various functions for CDF/PDF Graph Window. For more information, see Chapter 5_CDFPDF Menu_Map.pdf that comes with it.
Event	Change the properties of Event window. For more information, see Chapter 5_Event Menu_Map.pdf that comes with it.
Statistics	Change the properties of table or use the various functions for Statistics Tab Window. For more information, see Chapter 5_Event Menu_Map.pdf that comes with it.

Tree Mouse Menu

Tree Mouse Menu is displayed when you clicked right button of mouse while selecting the parameter tree. It is displayed differently depending on each parameter tree not tab.

For more information, see Chapter 5_Tree Mouse Menu.pdf that comes with it.

Contact Us



South Korea (HQ) : Innowireless Co., Ltd.

Innowireless B/D 190, Seohyeon-ro, Bundang-gu, Seongnam-si, Gyeonggi-do, South Korea

Tel :+82-31-788-1700 / Fax :+82-31-705-0712 salesteam@innowireless.com

http://www.innowireless.com

Hong Kong : Accuver APAC Limited

Unit 7, 21/F., Laws Commercial Plaza, No. 788 Cheung Sha Wan Road, Kowloon, Hong Kong SAR

Tel :+852-2-2210-7004 / Fax :+852-2-2210-7017 sales.apac@accuver.com

http://www.accuver.com

USA : Accuver Americas

500 N. Central Expressway Suite 210, Plano, TX, 75074, USA

Tel :+1-469-241-6100 / Fax :+1-469-241-6199 sales.usa@accuver.com support.usa@accuver.com

http://www.accuver.com http://help.usa.accuver.com

Japan : Accuver Co., Ltd.

23F Kamiyacho Trust Tower 4-1-1 Toranomon Minato-ku, Tokyo, Japan 105-6923

Tel :+81-3-6430-2580 / Fax :+81-3-6430-2589 sales@accuver.jp support_aj_1@accuver.com

http://www.accuver.jp

United Kingdom : Accuver EMEA Limited

Unit 20, Building 6, Croxley Green Business Park, Hatters Lane, Watford, WD18 8YH, United Kingdom

Tel:+44-203-457-4486 / Fax :+44-189-587-6338 sales.emea@accuver.com support.emea@accuver.com

http://www.accuver.com https://helpdesk.accuver.eu

Poland : Accuver EMEA sp. z o.o.

ul. Domaniewska 37, 02-672 Warszawa, Poland

Tel :+48-22-370-2518 sales.emea@accuver.com support.emea@accuver.com

http://www.accuver.com https://helpdesk.accuver.eu

India : Innowireless India Services Private Limited

Office No. 1304/1305, Maithili Signet, Plot No. 39/4, Sector 30-A, Vashi, Navi Mumbai, Maharashtra, India 400703

Tel :+91-224963-4902 / Fax :+91-222087-0178 innoindia.support@innowireless.com

http://www.accuver.com

China : Accuver Shanghai Co., Ltd.

Room 321, Building 1, 3152 HechuanRoad, MinhangDistrict, Shanghai, China

Tel : +86-187-0659-0330 sales.apac@accuver.com

http://www.accuver.com